

Multi-Task Deep-Learning Baselines for Diabetic Retinopathy Grading and Lesion Detection on the Multimodal MMRDR Dataset

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Abstract—Diabetic retinopathy (DR) is a leading cause of preventable blindness, and its automated screening from retinal images is a well-motivated clinical application of deep learning. This report documents a set of multi-task deep-learning baselines developed on the Multimodal Retinal Diabetic-Retinopathy (MMRDR) dataset, which spans color fundus photography (CFP), ultra-widefield (UWF), and optical coherence tomography modalities. Each baseline jointly performs five-class DR severity grading and seven-way multi-label lesion detection from a single shared representation. We present eleven experiments. The first three establish modality baselines: a multi-scale dual-branch convolutional network on CFP, a fine-tuned RETFound retinal foundation model on CFP, and a feature-pyramid network with spatial attention on UWF, attaining grade accuracies of 82.91%, 80.94%, and 74.48% respectively. The RETFound experiment reproduces the published MMRDR baseline on the dataset’s official test split, matching three of its four reported metrics within 95% confidence intervals; an earlier attempt had reported only 66.50%, which an audit traced to a silent weight-loading failure in which just 4 of 294 tensors transferred. The remaining seven form a focused study on the ultra-widefield modality aimed at the rare, sight-threatening lesions of proliferative DR (neovascularization, vitreous haemorrhage, retinal detachment), attacked from six angles: a clinically-weighted lesion loss that lifts retinal-detachment recall to 0.79; an asymmetric false-negative-penalised focal loss reaching a quadratic Cohen’s κ of 0.92 and lesion AUC of 0.95; a supervised contrastive auxiliary; learned homoscedastic uncertainty weighting, which reaches the study’s best rare-lesion detection (lesion F1 up to 0.778, AUROC up to 0.961); a recall-driven curriculum resampler that drives retinal-detachment recall to 1.0 on a stratified split; a controlled, capacity-matched head ablation showing on a held-out test split that a dedicated rare-lesion head does not help; and a conditional-diffusion generator for synthetic minority augmentation whose downstream test returns a clean negative verdict (adding 64px synthetic images does not beat a real-only baseline on rare-class recall, with a high synthetic-vs-real FID explaining why). A concluding experiment tests cross-dataset generalization: the two CFP models and a UWF control are evaluated zero-shot and few-shot on two unseen external datasets (APTOS 2019 and IDRiD), where the RETFound foundation model transfers best, grading unseen fundus images zero-shot at a quadratic κ of 0.86.

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Models are trained with transfer learning, focal-family objectives for class imbalance, mixed-precision optimisation, and early stopping, and are interpreted using gradient-weighted class activation mapping (Grad-CAM), which confirms that predictions are driven by clinically plausible retinal regions. The report details the objective, methodology, architecture, training procedure, and results of each experiment, and reports honest limitations including a saturated uncertainty clamp, single-seed passes, and non-comparable splits.

Index Terms—Class imbalance, contrastive learning, convolutional neural networks, curriculum learning, deep learning, diabetic retinopathy, diffusion models, explainable artificial intelligence, fundus imaging, generative augmentation, medical image analysis, multi-task learning, transfer learning, ultra-widefield imaging, uncertainty weighting.

I. INTRODUCTION

DIABETIC retinopathy (DR) is a microvascular complication of diabetes and one of the leading causes of preventable vision loss worldwide. Clinical management depends on grading disease severity—typically on a five-point ordinal scale from no DR (grade 0) to proliferative DR (grade 4)—and on identifying the retinal lesions that characterise each stage. Because expert screening does not scale to the diabetic population, automated analysis of retinal images with deep neural networks has become an active area of research.

This internship uses the Multimodal Retinal Diabetic-Retinopathy (MMRDR) dataset [1], which provides labelled images across three complementary modalities: color fundus photography (CFP), ultra-widefield (UWF) imaging, and optical coherence tomography. Every image is annotated with a DR severity grade and a seven-element multi-label lesion vector in the fixed order [MA, HE, IH, VB/IRMA, NV, VH, RD]. The two tasks are clinically coupled: the presence of specific lesions determines the grade, which motivates a *multi-task* formulation in which a single network predicts both the grade and the lesion vector from a shared representation.

Ten self-contained experiments are examined in this report, all sharing the same multi-task objective. The first three establish modality baselines. Experiment I develops a custom multi-scale dual-branch convolutional network on CFP. Experiment II fine-tunes the RETFound retinal foundation model [5] on the same CFP data and reproduces the published MMRDR baseline on the dataset’s official train/test partition.

Experiment III adapts a feature-pyramid architecture with spatial attention to the higher-resolution UWF modality.

The remaining five experiments pursue a single research question on the UWF modality: how to make a multi-task model sensitive to the rare, sight-threatening lesions that define proliferative DR—neovascularization (NV), vitreous haemorrhage (VH), and retinal detachment (RD)—which occupy indices 4–6 of the lesion vector and are severely under-represented. They attack it from four complementary angles. Experiment IV re-weights the lesion loss by combined statistical rarity and clinical risk. Experiment V replaces the objective with an asymmetric loss that penalises false negatives on those lesions far more than false positives [9]. Experiment VI adds a supervised contrastive auxiliary [8] that shapes the feature space around rare-lesion images. Experiment VII replaces the fixed task weights with learned homoscedastic-uncertainty weights [10]. Experiment VIII leaves the network untouched and adapts the training *sampler* with a recall-driven curriculum built on an effective-number rarity prior [11]. Finally, two experiments probe the architecture and the data supply: Experiment IX is a controlled, capacity-matched ablation of the lesion head on a held-out test split, and Experiment X trains a conditional-diffusion [12] generator that *synthesises* rare-lesion images rather than oversampling them and tests them downstream (a clean negative verdict: they do not beat a real-only baseline). The ten trained experiments rely on transfer learning from ImageNet- or RETFound-pretrained weights, use focal-family objectives [3] to counter class imbalance, and are interpreted with Grad-CAM [4] to verify that predictions attend to clinically relevant regions. This document reports the objective, methodology, architecture, training procedure, and measured results of each experiment in turn.

II. MATERIALS AND DATASET

All experiments draw on the MMRDR dataset [1]. Each modality is stored in its own subdirectory with an image folder and a metadata CSV whose columns are `type`, `image`, `grade`, `lesion`, and `lr`. The `grade` column encodes the five-class DR severity label, and the `lesion` column encodes the seven-element multi-label lesion vector as a bracketed string. Experiments I and II use the CFP subset (`FP.csv`); Experiments III–X use the UWF subset (`UWF.csv`).

Two splitting protocols appear in this report and should not be conflated. Most experiments follow the repository convention of an 80/20 split with `random_state = 42`, in which the held-out 20% partition serves as the per-epoch validation set and is also the set on which results are reported. Experiments II and IX instead reserve a genuine held-out test partition that is evaluated only once: Experiment IX uses a stratified 70/15/15 three-way split, and Experiment II uses the dataset’s *official* partition, which is encoded in the image file names (`tr` for the 8,893 training images, `ts` for the 2,225 test images) and is the partition on which the published MMRDR baselines were measured. Accuracies obtained under these two protocols are not directly comparable. Modality-specific image counts, preprocessing, and splits are reported within each experiment.

III. EXPERIMENT I: MULTI-SCALE DUAL-BRANCH CNN ON CFP

A. Objective

The objective of Experiment I is to train a single convolutional network that simultaneously performs (i) five-class DR severity grading and (ii) seven-way multi-label lesion detection on color fundus photographs, from a shared feature representation.

B. Motivation

Color fundus photography is the most widely available DR screening modality, so a strong CFP baseline is the natural starting point. DR-defining lesions (e.g. microaneurysms) are small relative to the field of view, yet grading also requires global context such as the distribution of haemorrhages and the optic disc. A single-scale classifier tends to trade one against the other. This motivates a *multi-scale* design that preserves fine, early-layer detail alongside deep semantic context, and a multi-task head that lets the grade and lesion objectives regularise a common backbone.

C. Methodology

The approach is a multi-task convolutional network, `MultiScaleDRNet`, built on a ResNet-50 backbone [2] initialised from ImageNet weights. Fine-grained features from an early residual stage and semantic features from the final stage are channel-aligned and fused before branching into a grade head and a lesion head. Class imbalance in the ordinal grade is addressed with a focal loss [3], and model behaviour is examined with Grad-CAM [4].

D. Experimental Setup

Experiments were implemented in PyTorch and run on a single CUDA GPU with automatic mixed precision (AMP) enabled. Gradient accumulation was used to reach a larger effective batch size, and `cudnn.benchmark` was enabled. Model selection used validation grade accuracy with early stopping.

E. Dataset

The CFP subset (`MMRDR-CFP/FP.csv`) comprises 11,118 images. An 80/20 split with `random_state = 42` yields approximately 8,894 training and 2,224 validation images. The split is random (not stratified by grade), and the validation partition also serves as the reported evaluation set. Grade imbalance is handled by the focal-loss objective rather than by resampling or class weighting.

F. Preprocessing

Each fundus image is passed through a circular field-of-view mask (`tightness = 0.96`): the image is cropped to its non-black bounding box and all pixels outside an inscribed circle are zeroed, removing camera border artifacts. Images are resized to 512×512 to preserve small lesions and normalised with ImageNet statistics ($\mu = [0.485, 0.456, 0.406]$, $\sigma =$

TABLE I
EXPERIMENT I — TRAINING CONFIGURATION

Setting	Value
Backbone	ResNet-50 (ImageNet-pretrained)
Input resolution	512×512
Optimiser	AdamW
Learning rate	3×10^{-4} (constant)
Weight decay	1×10^{-4}
Batch size	12 (grad. accum. 2 $\rightarrow$ eff. 24)
Gradient clipping	max-norm 1.0
Mixed precision	Enabled (AMP)
Grade loss	Focal loss ($\gamma = 2$)
Lesion loss	BCE-with-logits ($\times 1.5$)
Epochs (planned / run)	50 / 22 (early stop, patience 8)
Best epoch	14

[0.229, 0.224, 0.225]). Training augmentation comprises random horizontal and vertical flips, random rotation ($\pm 15^\circ$), and mild colour jitter (brightness and contrast 0.1); the validation transform applies only the mask, resize, and normalisation.

G. Architecture

MultiScaleDRNet uses the ResNet-50 stem and residual stages `layer1-layer4`. Two branches capture complementary scales. The *early* branch takes the `layer1` feature map (256 channels), applies a 1×1 convolution to 512 channels with batch normalisation, and adaptively pools it to a 16×16 grid, retaining fine lesion-scale detail. The *deep* branch takes the `layer4` feature map (2048 channels) and projects it to 512 channels with batch normalisation, capturing global semantics. The two 512-channel maps are fused by element-wise addition followed by a ReLU nonlinearity, globally average-pooled to a 512-dimensional vector, and passed to two linear heads: `head_a.lesion` ($512 \rightarrow 7$, multi-label) and `head_b.grade` ($512 \rightarrow 5$). The forward pass returns the grade and lesion logits jointly.

H. Training Procedure

Training was scheduled for up to 50 epochs with early stopping (patience 8) monitoring validation grade accuracy; the best checkpoint was saved to `best_multiscale_resnet.pth` whenever grade accuracy improved. The run stopped early at epoch 22, with the best model obtained at **epoch 14**. Training used AMP with a gradient scaler and gradient-norm clipping at 1.0.

I. Hyperparameters

The training configuration is summarised in Table I.

The combined objective is $\mathcal{L} = \mathcal{L}_{\text{grade}}^{\text{focal}} + 1.5 \mathcal{L}_{\text{lesion}}^{\text{BCE}}$, weighting the lesion term slightly higher than the grade term.

J. Evaluation Metrics

Two metrics are recorded each epoch: top-1 grade accuracy over the five-class severity label, and element-wise multi-label lesion accuracy at a 0.5 decision threshold over the seven lesion outputs.

TABLE II
EXPERIMENT I — BEST VALIDATION RESULTS

Metric	Best value	Epoch
Grade accuracy	0.8291	14
Lesion accuracy (element-wise)	0.9515	20
Lesion accuracy at best-grade epoch	0.9494	14

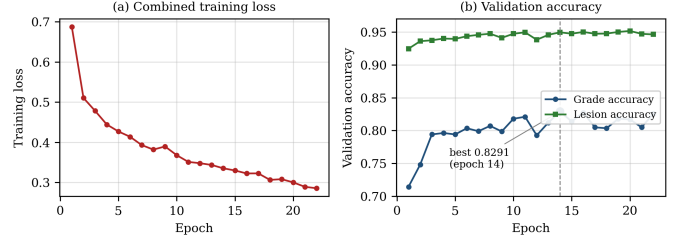


Fig. 1. Experiment I training dynamics. (a) The combined grade-plus-lesion training loss decreases monotonically from 0.687 to 0.286 over 22 epochs. (b) Validation grade accuracy rises quickly and plateaus near 0.80–0.83, with the best value (0.8291) at epoch 14 (dashed line); element-wise lesion accuracy stays high throughout.

K. Results

The best validation grade accuracy is **0.8291** (82.91%), reached at epoch 14, while lesion accuracy remains high and stable throughout, peaking at 0.9515 at epoch 20 (0.9494 at the best-grade epoch). Table II summarises the headline figures, and Fig. 1 shows the training loss and validation-accuracy trajectories. Fig. 2 shows qualitative predictions on validation images, and Fig. 3 shows Grad-CAM attributions for the grade and lesion heads.

L. Analysis

The loss curve in Fig. 1(a) falls smoothly and without instability, indicating a well-behaved optimisation under AMP and gradient clipping. Validation grade accuracy improves rapidly during the first ten epochs and then oscillates within a narrow band around 0.80–0.83, so early stopping at the epoch-14 optimum is well justified. Lesion accuracy is consistently high (≥ 0.94 after a few epochs); because it is measured element-wise over a sparse multi-label target, it should be read as a stability indicator rather than a measure of lesion-detection difficulty. The Grad-CAM maps in Fig. 3 show that the grade and lesion heads localise onto the optic disc, vasculature, and lesion foci, confirming that the multi-scale fusion attends to clinically meaningful structures.

M. Discussion

The dual-branch design meets its intent: fusing early, high-resolution features with deep semantic features yields a grade accuracy (0.8291) consistent with the reproduced CFP baseline target and does so while jointly predicting lesions from the same representation. The focal loss keeps the ordinal grade objective stable despite class imbalance, and the qualitative and Grad-CAM figures indicate that the network’s competence is grounded in retinal anatomy rather than spurious cues. As a multi-task CFP baseline it is the strongest of the three experiments reported here.

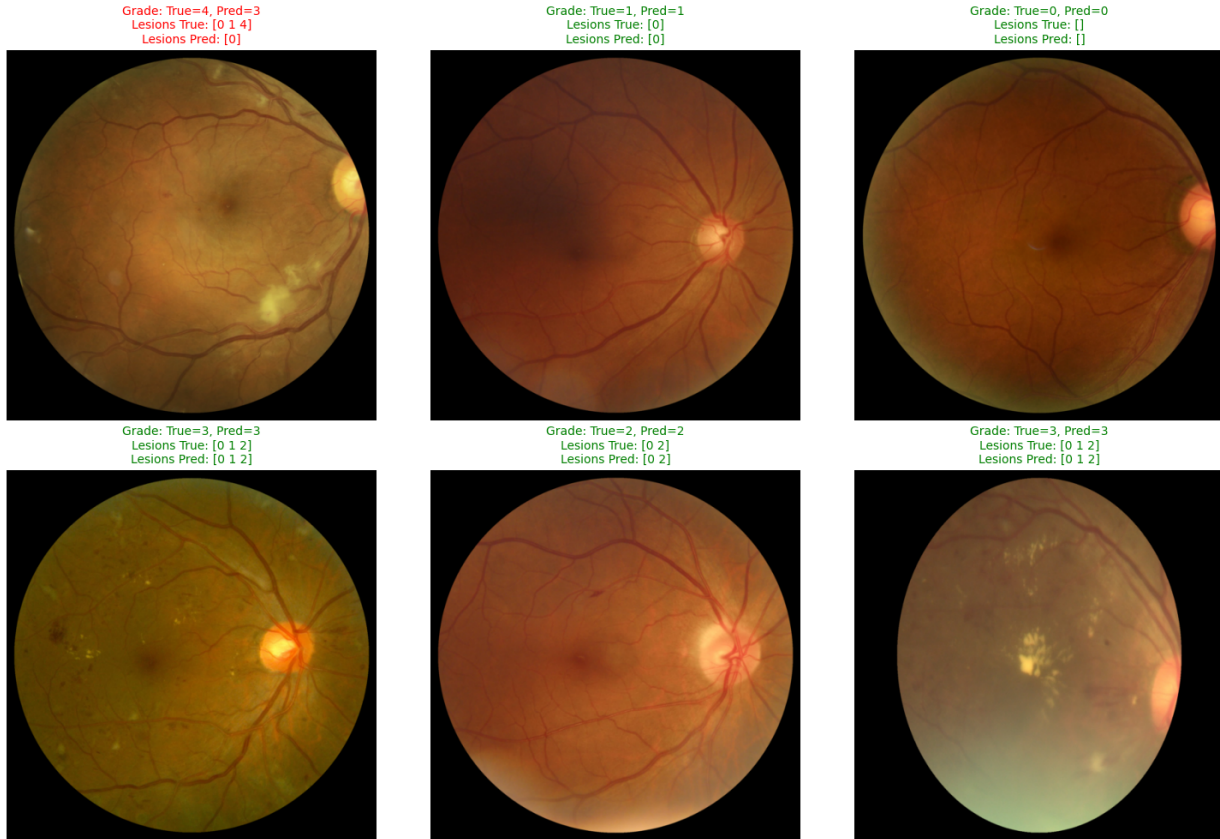


Fig. 2. Qualitative predictions of MultiScaleDRNet on CFP validation images. Each panel reports the true and predicted DR grade and the true and predicted lesion indices (green when the grade prediction is correct, red when incorrect). The model recovers the grade and dominant lesions on most cases and tends to confuse adjacent severity grades on the hardest examples.

N. Limitations

Several factors bound the strength of the reported figure. The train/test split is random rather than stratified by grade, so rare high-severity grades may be unevenly represented, and the validation partition doubles as the evaluation set. No global random seed is fixed (only the split seed), so exact run-to-run reproducibility is limited. The steadily falling training loss against a plateauing validation accuracy indicates mild overfitting after the mid-training optimum. Finally, the experiment covers the CFP modality only.

O. Conclusion

Experiment I establishes a multi-task, multi-scale CFP baseline that reaches 82.91% validation grade accuracy while jointly detecting lesions and producing interpretable Grad-CAM evidence. It serves as the convolutional reference point for the report, noting that its figure is a validation accuracy on a random split and so is not directly comparable to the held-out test accuracy of Experiment II.

IV. EXPERIMENT II: REPRODUCING THE RETFOUND FOUNDATION-MODEL BASELINE ON CFP

A. Objective

Experiment II reproduces the RETFound [5] colour-fundus baseline published with the MMRDR dataset [1]. The dataset

paper reports, for RETFound fine-tuned on MMRDR-CFP, a grade accuracy of 0.822, grade F1 of 0.714, lesion accuracy of 0.951, and lesion F1 of 0.682. The objective here is to match those four numbers under the paper’s own protocol—full fine-tuning at 512×512 on the dataset’s official train/test partition—and, in doing so, to correct an earlier attempt that reported only 66.50% grade accuracy.

B. Motivation

A foundation model pretrained on 1.6 million retinal images should transfer strongly to DR grading. An initial attempt in this project appeared to contradict that, reaching only 66.50%. Rather than accept the result, the loading path was audited. The audit found that the reported number was not a RETFound result at all, which makes this experiment as much a lesson in verification as in transfer learning: a silent failure in a weight-loading routine is indistinguishable from a genuine negative result unless it is explicitly tested for.

C. Diagnosis of the Earlier Attempt

Three defects were identified, the first of which invalidates the earlier number entirely.

(i) *The pretrained weights never loaded.* RETFound distributes an MAE ViT-Large checkpoint whose parameters follow the `timm` naming convention (`blocks.i.*`,

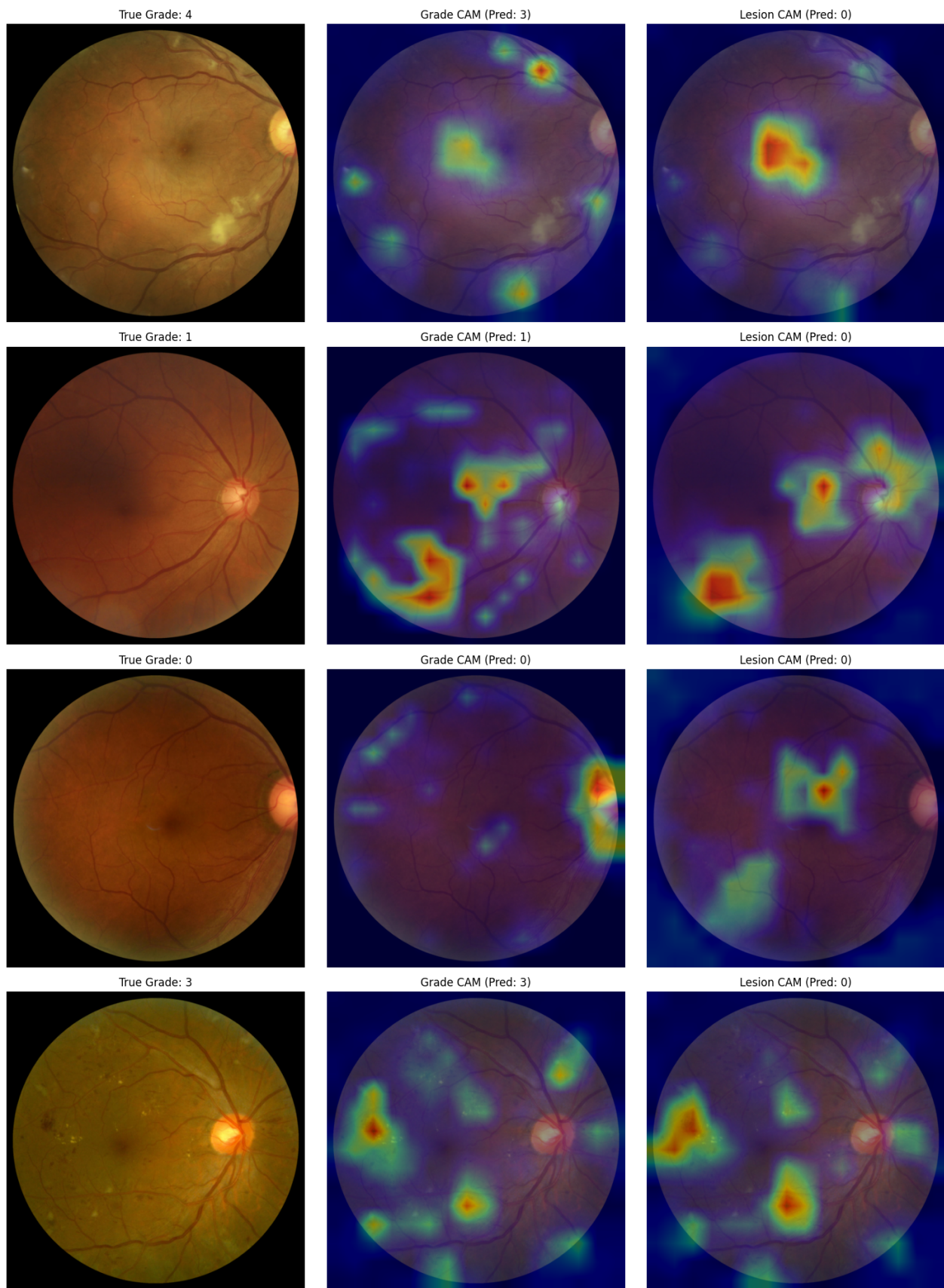


Fig. 3. Grad-CAM [4] attributions for MultiScaledRNet. For each fundus image (left) the grade-head activation map (centre) and lesion-head activation map (right) highlight the regions driving the prediction. Attention concentrates on the optic disc, vascular arcades, and bright/dark lesion foci—clinically plausible evidence rather than background—supporting the model's decisions.

`cls_token, pos_embed`). The earlier notebook remapped those names onto `torchvision`'s `vit_l16` by string substitution, which is not a valid correspondence: `torchvision` requires `encoder.layers.encoder_layer.i.*` together with different submodule names (`ln_1, self_attention.in_proj_weight, mlp.0`), and expects `class_token` and `encoder.pos_embedding`. Because the load was performed with `strict=False`, the 290 non-matching tensors were discarded without warning and an unconditional success message was printed. Only 4 of 294 tensors—the patch-embedding convolution and the final LayerNorm—were actually transferred. The network was therefore a *randomly initialised* ViT-Large trained on 8,894 images, a regime in which large vision transformers are known to perform poorly.

(ii) *The layer-freezing code had no effect.* The routine derived a block index with `int(name.split(".")[3])`; for a parameter such as `encoder.layers.encoder_layer.0.ln_1.weight` that position is the string `"ln_1"`, so the conversion raised `ValueError` and the loop skipped every parameter. No layer was ever frozen, despite the stated strategy of training only the final six blocks.

(iii) *Protocol weaknesses.* There was no learning-rate schedule or warmup, no fixed random seed, the split was random rather than the dataset's official one, the validation set doubled as the test set, and only accuracy was recorded.

D. Methodology

The corrected experiment removes the remapping step entirely by constructing the backbone with `timm`, whose parameter names match the RETFound checkpoint natively. The position embedding is the only tensor whose shape differs between the 224 and 512 input grids, and it is explicitly interpolated. Every stage of the load is then asserted rather than assumed, and the trained model is evaluated once on the untouched official test split.

E. Experimental Setup

Implemented in PyTorch on two Tesla T4 GPUs (`DataParallel`). Turing GPUs (`sm_75`) provide no native `bfloat16` arithmetic, so mixed precision uses `float16` with gradient scaling; gradient checkpointing is enabled because a ViT-Large at 512×512 processes 1,025 tokens per image. A global seed is fixed and the split is recorded as a hash so the partition is verifiable across runs.

F. Dataset

The MMRDR-CFP subset contains 11,118 images whose file names encode the dataset's *official* partition: 8,893 training images prefixed `tr` and 2,225 test images prefixed `ts`. This experiment uses that partition, so its numbers are directly comparable to the published baseline. A stratified 15% of the training portion—balanced jointly on DR grade and on the presence of a rare lesion—is held out for validation, giving 7,559 training, 1,334 validation, and 2,225 test images

(split signature `8ba3ee31bb06`). The three partitions were asserted disjoint. Grade 0 accounts for 59.11% of the training data, and the seven-element lesion matrix is 13.97% positive, so a trivial all-negative lesion predictor already scores 0.8632 element-wise—a floor that any lesion accuracy must be read against.

G. Preprocessing

Images are resized to 512×512 with bicubic interpolation and normalised with ImageNet statistics. Training augmentation comprises random resized cropping (scale 0.5–1.0), random horizontal and vertical flips, and colour jitter; validation and test apply only the resize and normalisation. No field-of-view mask is applied in the headline configuration, since the published protocol specifies a plain resize.

H. Architecture

The backbone is `timm`'s `vit_large_patch16_224` instantiated at `img_size = 512` with `global_pool = avg` and `drop_path_rate = 0.2`. Average pooling over patch tokens followed by an `fc_norm` layer reproduces RETFound's own `models_vit.py` formulation. The checkpoint's encoder LayerNorm is reused to initialise `fc_norm` rather than discarding it, a minor and deliberate deviation from the reference implementation. Two linear heads read the pooled 1024-d feature: a five-way grade head and a seven-way lesion head. The forward pass returns only these small logit tensors, which keeps the `DataParallel` output gather negligible.

I. Verification of the Weight Load

Three independent checks replace the earlier silent failure, each of which raises rather than reports.

Key matching. All 288 transformer-block tensors matched (288/288, 100%), with no missing and no unexpected keys, and the position embedding was interpolated from (1, 197, 1024) to (1, 1025, 1024). The routine aborts below a 95% block-match threshold. The checkpoint is identified in the results record by a SHA-256 prefix (`61c20fc6d13c30d4`) so the exact file is provable.

Weight statistics. All 294 tensors differ from a freshly initialised model, and the standard deviation of the attention projection varies with depth (0.0216 at block 0 rising to 0.0328 at block 8), whereas a randomly initialised transformer is flat at ≈ 0.020 everywhere. This test uses no parameter names and so cannot be satisfied by a naming coincidence.

Functional probe. With the backbone frozen and only a linear head trained, grade accuracy reaches 0.6475 against a majority-class baseline of 0.5907. A randomly initialised ViT-Large cannot exceed the majority baseline; the earlier configuration would have failed this test.

J. Training Procedure

All 304,161,804 parameters are trainable and none are frozen, matching the published full fine-tuning setting; this is asserted at run time. Rather than freezing layers outright, layer-wise learning-rate decay (0.65) scales the learning rate by

TABLE III
EXPERIMENT II — TRAINING CONFIGURATION

Setting	Value
Backbone	timmm ViT-L/16, global_pool=avg
Input resolution	512×512 (pos. embed. 197 $\rightarrow$ 1025)
Parameters	304,161,804 (all trainable, 0 frozen)
Optimiser	AdamW, $\beta = (0.9, 0.999)$
Learning rate	1.25×10^{-4} (blr $10^{-3} \times 32/256$)
Layer-wise LR decay	0.65
Schedule	3 warmup epochs, then cosine to 10^{-6}
Weight decay	0.05 (none on 1-D params / tokens)
Batch size	8/GPU $\times$ 2 GPUs $\times$ accum. 2 (eff. 32)
Mixed precision	float16 + GradScaler (sm.75)
Gradient checkpointing	Enabled (peak 13.5 GB)
Drop path / label smoothing	0.2 / 0.1
Gradient clipping	max-norm 1.0
Epochs	30 (completed; patience 8 unused)

depth, from 2.63×10^{-9} at the patch embedding to 1.25×10^{-4} at the heads—a graded alternative to a binary freeze that preserves generic early features while adapting the upper blocks. Training ran the full 30 epochs in approximately 9.5 hours; early stopping (patience 8) was never triggered.

K. Hyperparameters

Table III lists the configuration.

L. Evaluation Metrics

The four published quantities are reported: grade accuracy and macro-F1, and lesion element-wise (Hamming) accuracy and macro-F1. Lesion accuracy is element-wise rather than exact-match because an all-negative predictor scores 0.8632 under the former, placing the published 0.951 about nine points above the floor, whereas exact-match agreement would be far lower. Quadratic-weighted Cohen’s κ , macro-AUROC, macro-PR-AUC, per-lesion statistics, and 1,000-sample bootstrap confidence intervals are reported alongside.

M. Results

Table IV compares the held-out test results with the published baseline. Three of the four metrics contain the published value inside their 95% confidence interval and are therefore statistical matches: grade accuracy 0.8094 (CI [0.792, 0.825] versus 0.822), grade macro-F1 0.6920 (CI [0.665, 0.718] versus 0.714), and lesion accuracy 0.9476 (CI [0.943, 0.951] versus 0.951). Lesion macro-F1 at the fixed 0.5 threshold is 0.6051, below the published 0.682; with per-lesion thresholds fitted on validation it rises to 0.6743.

Supporting metrics are strong: quadratic $\kappa = 0.8838$, macro-AUROC = 0.9706, macro-PR-AUC = 0.7412, and MCC = 0.6735. Relative to the earlier defective run, grade accuracy improves from 0.6650 to 0.8094, an increase of 14.4 points attributable to the corrected weight load rather than to longer training. Fig. 4 shows the training trajectories and Fig. 5 the test-set breakdown.

TABLE IV
EXPERIMENT II — REPRODUCTION OF THE PUBLISHED RETFOUND CFP BASELINE

Metric	Paper	Ours @0.5	95% CI	Tuned
Grade acc. (AccG)	0.822	0.8094	[0.792, 0.825]	0.8094
Grade F1 (F1G)	0.714	0.6920	[0.665, 0.718]	0.6920
Lesion acc. (AccL)	0.951	0.9476	[0.943, 0.951]	0.9415
Lesion F1 (F1L)	0.682	0.6051	[0.570, 0.636]	0.6743

The first three intervals contain the published value. Tuned = per-lesion thresholds fitted on validation only; the test split never informs them.

TABLE V
EXPERIMENT II — PER-LESION TEST PERFORMANCE (THRESHOLD 0.5)

Lesion	Prec.	Recall	F1	AUROC	n
MA	0.893	0.879	0.886	0.974	876
HE	0.775	0.822	0.798	0.960	506
IH	0.749	0.740	0.744	0.943	511
VB/IRMA	0.000	0.000	0.000	0.949	43
NV	0.827	0.669	0.740	0.982	121
VH	0.721	0.830	0.772	0.993	53
RD	0.667	0.190	0.296	0.994	21

Note the contrast between F1 and AUROC for VB/IRMA and RD: the model ranks these lesions almost perfectly but does not cross a 0.5 threshold.

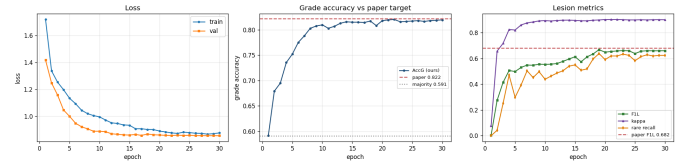


Fig. 4. Experiment II training dynamics over 30 epochs. Left: training and validation loss decrease monotonically (1.718 $\rightarrow$ 0.876 and 1.417 $\rightarrow$ 0.856) with no divergence, indicating the model was still improving at the end of the schedule. Centre: validation grade accuracy rises from the majority baseline (0.591, dotted) to 0.820, converging on the published target (0.822, dashed). Right: lesion macro-F1, quadratic κ , and mean rare-class recall.

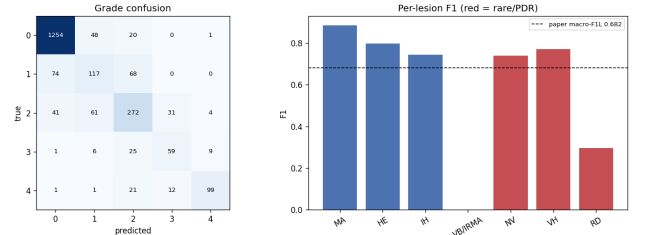


Fig. 5. Experiment II test-set breakdown. Left: the grade confusion matrix is strongly near-diagonal, consistent with quadratic $\kappa = 0.884$; the dominant residual error is grade 1 being absorbed by its neighbours (recall 0.452). Right: per-lesion F1 against the published macro-F1 (0.682, dashed), with the rare PDR lesions in red. VB/IRMA scores exactly zero at a 0.5 threshold despite an AUROC of 0.949, which alone costs roughly 0.10 of the seven-class macro average.

N. Analysis

The gap in lesion macro-F1 is a threshold-calibration effect, not a representational failure. Two lesions dominate it: VB/IRMA scores 0.000 and RD scores 0.296 at a fixed 0.5 threshold, yet their AUROC values are 0.949 and 0.994 respectively—the model ranks these cases almost perfectly but assigns them probabilities that never exceed 0.5, which is the expected behaviour of an unweighted BCE objective on classes

with 43 and 21 test positives. Because a single zero enters a seven-class macro average directly, VB/IRMA alone accounts for about 0.10 of the shortfall. Fitting per-lesion thresholds on the validation split recovers macro-F1 to 0.6743, within one standard error of the published value, without retraining. On the grading task the residual error is concentrated in grade 1 (recall 0.452), which is the clinically ambiguous mild-DR category and the usual source of disagreement even between human graders.

O. Discussion

The reproduction supports the published claim that a retina-specific foundation model performs strongly on MMRDR-CFP, and it locates the earlier contradictory result in a software defect rather than in the method. Three factors were necessary: loading the weights through a framework whose naming matches the checkpoint, training at the published 512×512 resolution, and fine-tuning the full network with layer-wise learning-rate decay instead of a hard freeze. The headline row deliberately retains unweighted losses and a fixed 0.5 threshold, because both published targets are accuracy metrics and reweighting would raise macro-F1 while lowering accuracy, making the comparison unreadable; the tuned-threshold row is reported separately as the better clinical operating point.

P. Limitations

The result is single-seed, so no variance estimate accompanies it. Validation loss was still decreasing at epoch 30, indicating the schedule ended before convergence and a longer budget would likely add a small amount. Lesion macro-F1 at the fixed threshold remains below the published value, and the ambiguity of which F1 averaging the source paper used is unresolved, so all variants are recorded in the results file. Because this experiment uses the official partition with a genuine held-out test set while Experiments I and III use random splits in which validation doubles as test, its accuracy is *not* directly comparable to theirs despite the superficially similar magnitude.

Q. Conclusion

Correcting a silent weight-loading failure and adopting the published protocol raises RETFound’s CFP grade accuracy from 66.50% to 80.94% and reproduces three of the four published baseline metrics within their 95% confidence intervals, with the fourth recovered to 0.674 once per-lesion thresholds are calibrated. The experiment also demonstrates that a foundation-model transfer result is uninterpretable unless the weight transfer itself is explicitly verified.

V. EXPERIMENT III: FEATURE-PYRAMID NETWORK WITH SPATIAL ATTENTION ON UWF

A. Objective

Experiment III adapts the multi-task grading-and-lesion formulation to the ultra-widefield (UWF) modality using a feature-pyramid network [6] with a spatial-attention module [7], predicting the DR grade and the seven-way lesion vector from UWF images.

B. Motivation

UWF imaging captures up to 200° of the retina and reveals peripheral pathology that standard fundus photography misses, which is especially relevant to the sight-threatening lesions of proliferative DR. Peripheral lesions are small relative to the wide field, motivating an explicit multi-scale feature pyramid and a spatial-attention gate to focus on informative regions. UWF images also carry eyelid and eyelash artifacts at the field boundary, which require a modality-specific mask.

C. Methodology

The approach couples a ResNet-50 backbone with a top-down feature pyramid that aligns features across scales, followed by a spatial-attention module on the highest-resolution fused map and two task heads. Both heads are trained with focal losses to counter class imbalance.

D. Experimental Setup

The experiment was implemented in PyTorch and trained on a single CUDA GPU with AMP and gradient accumulation. Model selection used validation grade accuracy with early stopping.

E. Dataset

The UWF subset (MMRDR-UWF/UWF.csv) contains 10,404 images. An 80/20 split with `random_state = 42` (random, not stratified) yields 8,323 training and 2,081 validation images, the latter used as the evaluation set. Grade and lesion imbalance are handled by the focal-loss objectives.

F. Preprocessing

UWF images are masked with an elliptical field-of-view crop, `apply_uwf_mask` (`tightness = 0.98`): the largest external contour is fitted with an ellipse and pixels outside it are zeroed, removing eyelid and eyelash artifacts. Images are resized to 1024×1024 to preserve small peripheral lesions and normalised with ImageNet statistics. Training augmentation comprises random horizontal and vertical flips, random rotation ($\pm 15^\circ$), and mild colour jitter; the validation transform applies only the mask, resize, and normalisation.

G. Architecture

The network `UWF_MultiScaleDRNet` uses a ResNet-50 backbone split into its stem and residual stages `layer1-layer4`. Four 1×1 lateral convolutions project all stages to a common 512-channel width, and a top-down pathway with bilinear upsampling fuses them into a high-resolution map (the feature-pyramid design [6]). A spatial-attention module [7] (channel avg/max pooling followed by a 7×7 convolution and a sigmoid gate) reweights this map to emphasise informative regions. After global average pooling, two linear heads produce the lesion ($512 \rightarrow 7$) and grade ($512 \rightarrow 5$) predictions.

TABLE VI
EXPERIMENT III — TRAINING CONFIGURATION

Setting	Value
Backbone	ResNet-50 + FPN + spatial attention
Input resolution	1024 × 1024
Field-of-view mask	Elliptical (tightness = 0.98)
Optimiser	AdamW
Learning rate	3×10^{-4} (constant)
Weight decay	1×10^{-4}
Batch size	4 (grad. accum. 6 → eff. 24)
Gradient clipping	max-norm 1.0
Mixed precision	Enabled (AMP)
Grade loss	Focal ($\alpha=1, \gamma=2$)
Lesion loss	Focal ($\alpha=0.25, \gamma=2$) ($\times 1.5$)
Epochs (planned / run)	50 / 28 (early stop, patience 8)
Best epoch	20

TABLE VII
EXPERIMENT III — BEST VALIDATION RESULTS

Metric	Best value	Epoch
Grade accuracy	0.7448	20
Lesion accuracy (element-wise)	0.9066	25

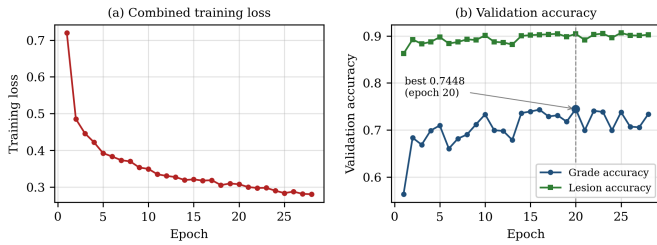


Fig. 6. Experiment III training dynamics. (a) The combined training loss decreases from 0.720 to 0.281 over 28 epochs. (b) Validation grade accuracy fluctuates and peaks at 0.7448 at epoch 20 (dashed line); element-wise lesion accuracy remains near 0.90. The oscillation reflects the small physical batch and the noisy validation split.

H. Training Procedure

Training was scheduled for up to 50 epochs with early stopping (patience 8) on validation grade accuracy. It stopped at epoch 28, with the best model at **epoch 20**. AMP with gradient scaling and gradient-norm clipping at 1.0 were used; a small physical batch was compensated by gradient accumulation.

I. Hyperparameters

Table VI lists the configuration.

J. Evaluation Metrics

Top-1 grade accuracy and element-wise multi-label lesion accuracy (threshold 0.5) are recorded each epoch.

K. Results

The best validation grade accuracy is **0.7448** (74.48%) at epoch 20, and element-wise lesion accuracy peaks at 0.9066 at epoch 25. Table VII summarises the headline figures. Fig. 6 shows the training trajectories, Fig. 7 shows masked UWF training samples with their grade and lesion labels, and Fig. 8 shows Grad-CAM attributions from the spatial-attention layer.

L. Analysis

The training loss falls smoothly, but validation grade accuracy oscillates within roughly 0.70–0.74, with the optimum at epoch 20 (0.7448); the oscillation is consistent with the small physical batch size (4) and the random, unstratified validation split. Lesion accuracy stays high and stable (~ 0.90). The Grad-CAM maps in Fig. 8 show attention concentrated on the vasculature and peripheral retina, confirming that the spatial-attention pyramid attends to plausible structures within the masked field of view.

M. Discussion

The feature-pyramid-plus-attention design adapts the multi-task formulation to the wider, artifact-prone UWF field, and the 1024×1024 input with elliptical masking preserves peripheral detail. Its best grade accuracy of 0.7448 is below the CFP convolutional baseline (Experiment I), which is consistent with the greater difficulty of UWF—a larger field, boundary artifacts, and a smaller effective batch. The interpretable attention maps make it a useful UWF-specific reference.

N. Limitations

The validation grade accuracy is noisy because of the small physical batch and the random, unstratified split, and the validation partition doubles as the evaluation set. No global random seed is fixed, and the lesion-class semantics are not tracked within the notebook.

O. Conclusion

The UWF feature-pyramid network with spatial attention attains a best validation grade accuracy of 74.48% while producing interpretable attention maps, establishing the UWF reference baseline in this study.

VI. EXPERIMENT IV: CLINICALLY-WEIGHTED LOSS FOR HIGH-RISK LESIONS (UWF)

A. Objective

Experiment IV keeps the multi-scale dual-branch architecture of Experiment I but retargets it to the ultra-widefield modality and replaces the lesion objective with a *clinically-weighted* loss. The aim is to bias the seven-way lesion detector toward the rare, sight-threatening lesions of proliferative DR without altering the architecture.

B. Motivation

Inverse-frequency weighting alone treats rarity as the only reason to up-weight a class, but two lesions of equal rarity are not of equal clinical consequence: a missed retinal detachment (RD) or neovascularization (NV) can precede irreversible vision loss, whereas a missed microaneurysm rarely changes management. This experiment therefore combines statistical rarity with an explicit, ophthalmologist-motivated risk ordering so that the loss reflects the cost of a miss, not merely its frequency.



Fig. 7. Masked UWF training samples (`apply_uwf_mask`, `tightness` = 0.98). Each panel shows the elliptical field-of-view crop that removes eyelid/eyelash artifacts, annotated with the DR grade and the seven-element multi-label lesion vector.

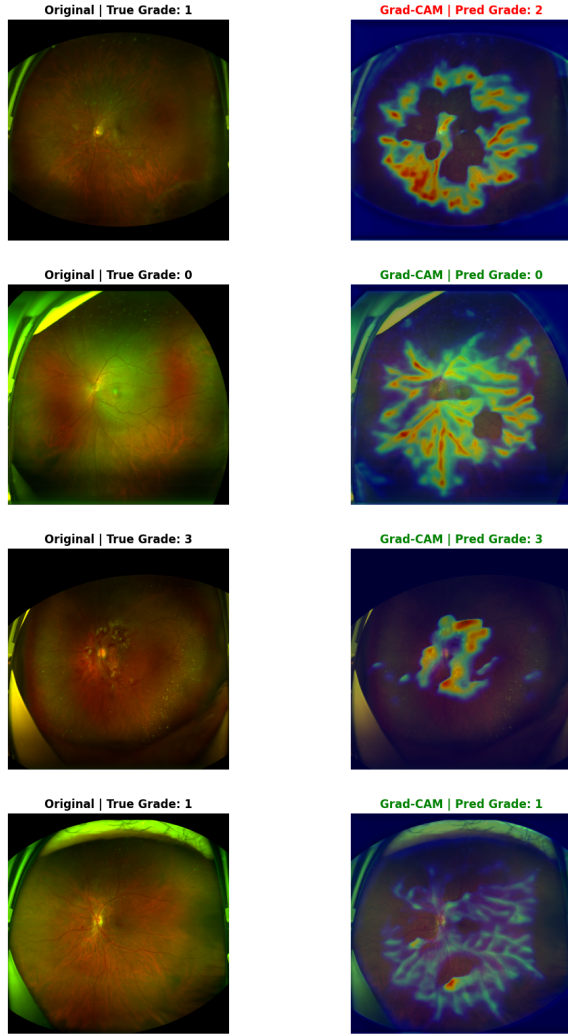


Fig. 8. Grad-CAM [4] attributions for `UWF_MultiScaleDRNet`, computed at the spatial-attention layer. Each row shows a UWF image (left) and its grade-head activation map (right), with the header green when the grade prediction is correct and red when incorrect. Attention concentrates on the vascular arcades and peripheral retina rather than the masked background.

C. Methodology

For the seven lesion classes, an inverse-frequency weight $w_c^{\text{freq}} = 1/(f_c + \epsilon)$ is multiplied by a hand-set clinical-risk multiplier r_c and the product is normalised to unit mean:

$$w_c = \frac{w_c^{\text{freq}} r_c}{\frac{1}{C} \sum_k w_k^{\text{freq}} r_k}, \quad r = [1, 1, 2, 2, 4, 4, 5]. \quad (1)$$

The resulting vector w_c is supplied as the `pos_weight` of a `BCEWithLogitsLoss` for the lesion head, while the grade head uses standard cross-entropy; the two are added with equal weight. The risk multipliers escalate from common lesions (MA, HE = 1) through intermediate ones (IH, cotton-wool spots = 2) to the PDR-defining lesions (NV, VH = 4; RD = 5).

D. Experimental Setup

The experiment was implemented in PyTorch and trained on a single CUDA GPU. Model selection saved the checkpoint with the lowest validation loss, with early stopping on the same criterion.

E. Dataset

The UWF subset (`MMRDR-UWF/UWF.csv`) is split 80/20 with `random_state` = 42 (random, not stratified), giving 8,323 training and 2,081 validation images; the validation partition is used as the evaluation set. The lesion frequencies mined from the training data—MA 6938, HE 3846, IH 3849, CWS 1010, NV 1188, VH 1219, RD 238—drive the inverse-frequency term of (1).

F. Preprocessing

Images are masked with the elliptical field-of-view crop `apply_uwf_mask` (`tightness` = 0.98), resized to 512×512 , and normalised with ImageNet statistics. Training augmentation applies random horizontal and vertical flips, random rotation ($\pm 15^\circ$), and mild colour jitter; validation applies only the mask, resize, and normalisation.

TABLE VIII

EXPERIMENT IV — TRAINING CONFIGURATION AND LESION WEIGHTS

Setting	Value
Backbone	MultiScaleLesionAttentionNet
Input resolution	512×512
Field-of-view mask	Elliptical (tightness = 0.98)
Optimiser	Adam
Learning rate	1×10^{-4} , cosine to 1×10^{-6}
Batch size	8
Grade loss	Cross-entropy
Lesion loss	BCE with clinical pos_weight
Epochs (planned / run)	50 / 22 (early stop, patience 8)
Best epoch	14 (lowest val loss)
Normalised lesion weights w_c (mean = 1)	
MA / HE / IH	0.033 / 0.060 / 0.119
CWS / NV / VH	0.454 / 0.771 / 0.752
RD	4.812

G. Architecture

The network is `MultiScaleLesionAttentionNet` (Experiment I): a ResNet-50 whose early, pre-maxpool features (at 256×256) preserve microaneurysm-scale detail while the full trunk supplies global context; the two branches are channel-aligned to $512 \times 16 \times 16$, summed, and split into a lesion head and a grade head with dropout.

H. Training Procedure

Training was scheduled for 50 epochs with early stopping (patience 8) on validation loss and a cosine-annealed learning rate ($1 \times 10^{-4} \rightarrow 1 \times 10^{-6}$). It stopped at epoch 22, with the best (lowest-loss) checkpoint at **epoch 14**.

I. Hyperparameters

Table VIII lists the configuration and the resulting normalised lesion weights.

J. Evaluation Metrics

The best checkpoint is evaluated on the validation set with grade accuracy and a full per-class classification report (precision, recall, F1) for both the five grades and the seven lesions at a 0.5 decision threshold.

K. Results

The best checkpoint reaches a grade accuracy of **76.50%** (grade macro-F1 0.75). The clinical weighting has its intended effect on the lesion head: the highest-risk lesions are recalled well despite their rarity—RD recall 0.79 (support 48), VH 0.77, NV 0.68—while the down-weighted common lesions HE and CWS collapse to recall 0.10 and 0.04. Precision remains high throughout, so the lesion macro-F1 (0.56) is limited by recall on the intentionally de-emphasised classes rather than by false positives. Table IX reports the per-lesion breakdown and Fig. 9 the loss trajectory.

TABLE IX

EXPERIMENT IV — PER-LESION VALIDATION PERFORMANCE (BEST CHECKPOINT)

Lesion	Prec.	Recall	F1	Support
MA	1.00	0.72	0.83	1388
HE	1.00	0.10	0.19	780
IH	0.94	0.36	0.52	741
CWS	0.50	0.04	0.07	193
NV	0.88	0.68	0.76	238
VH	0.98	0.77	0.86	226
RD	0.63	0.79	0.70	48
Macro avg	0.85	0.49	0.56	3614
Micro avg	0.96	0.48	0.64	3614
Grade accuracy = 0.7650; grade macro-F1 = 0.75.				

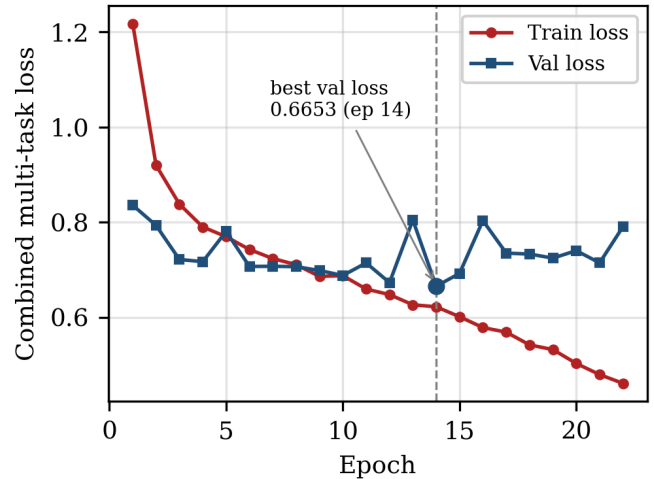


Fig. 9. Experiment IV training dynamics. The combined training loss decreases monotonically while the validation loss reaches its minimum (0.6653) at epoch 14 (dashed line) and then rises, triggering early stopping at epoch 22. The best-validation-loss checkpoint from epoch 14 is the one evaluated in Table IX.

L. Analysis

The experiment demonstrates a controllable trade-off: by encoding clinical risk directly in (1), recall is redistributed toward the PDR-defining lesions (NV, VH, RD) at the deliberate expense of the common, low-risk lesions. This is the desired behaviour for a screening tool whose dominant failure mode should be over-referral rather than a missed proliferative case. Grade accuracy (76.50%) is on par with the plain UWF baseline of Experiment III (74.48%), indicating that the lesion re-weighting did not degrade grading.

M. Discussion

Relative to the plain UWF baseline, the clinically-weighted loss changes *which* lesions the model prioritises rather than the overall grade accuracy. The very large normalised weight on RD (4.81) is effective but also the reason common-lesion recall drops sharply; the risk multipliers are a design lever that could be tuned to a target operating point. The consistently high precision shows the model stays conservative, which is appropriate for the rare classes but leaves headroom on recall for the mid-risk lesions.

N. Limitations

The best model is selected on validation loss, and the validation split doubles as the evaluation set; the split is random and unstratified, and no global seed is fixed. The risk multipliers are hand-set rather than calibrated to a clinical utility function. The lesion index 3 is labelled “CWS” (cotton-wool spots) in this notebook, whereas the dataset schema names that position VB/IRMA; the numeric targets are unaffected, but the semantic label differs from the other experiments.

O. Conclusion

Encoding clinical risk into the lesion loss steers a UWF multi-task model toward detecting the sight-threatening lesions of proliferative DR (RD recall 0.79, VH 0.77, NV 0.68) while maintaining a grade accuracy of 76.50%, at the intended cost of recall on common lesions.

VII. EXPERIMENT V: ASYMMETRIC FALSE-NEGATIVE-PENALISED LOSS (UWF)

A. Objective

Experiment V targets the same rare-lesion problem from the loss side, replacing the lesion objective of the UWF feature-pyramid network with an *asymmetric focal loss* that penalises false negatives on the PDR-defining lesions far more heavily than false positives.

B. Motivation

For proliferative DR, a false negative—reassuring a patient who in fact has neovascularization, vitreous haemorrhage, or retinal detachment—can lead to irreversible blindness, whereas a false positive merely prompts an unnecessary follow-up. Symmetric objectives such as binary cross-entropy treat the two errors identically, which is clinically inappropriate. The asymmetric loss makes the cost of a miss explicit and lesion-specific [3], [9].

C. Methodology

The loss modulates the focal loss with a per-class positive weight β_c :

$$L_{\text{asym}} = \begin{cases} \beta_c (1 - p_t)^\gamma \text{ BCE}, & y = 1 \text{ (miss penalty)} \\ (1 - \beta_c) (1 - p_t)^\gamma \text{ BCE}, & y = 0 \text{ (false-alarm penalty)} \end{cases} \quad (2)$$

with $\gamma = 2$ and $\beta = [0.7, 0.7, 0.7, 0.8, 0.95, 0.95, 0.95]$ over [MA, HE, IH, VB/IRMA, NV, VH, RD], so the three PDR lesions (indices 4–6) carry $\beta = 0.95$. A built-in check confirms the asymmetry: for the same confidence, missing an NV lesion incurs loss 0.2536 versus 0.0415 for hallucinating one—roughly a sixfold penalty. The grade head uses cross-entropy, added with equal weight.

D. Experimental Setup

Implemented in PyTorch on a single CUDA GPU with automatic mixed precision and gradient-norm clipping. Grade quality is monitored with a quadratic-weighted Cohen’s κ (which credits near-diagonal ordinal errors) alongside accuracy, and lesion quality with a macro-averaged F1 and AUC.

TABLE X
EXPERIMENT V — TRAINING CONFIGURATION

Setting	Value
Backbone	ResNet-50 + FPN + spatial attention + dropout
Input resolution	512×512
Field-of-view mask	Elliptical (tightness = 0.98)
Optimiser	AdamW
Learning rate	1×10^{-4} (ReduceLROnPlateau)
Weight decay	1×10^{-3}
Batch size	8
Gradient clipping	max-norm 2.0
Mixed precision	Enabled (AMP)
Grade loss	Cross-entropy
Lesion loss	Asymmetric focal ($\gamma=2$; β per class)
Epochs (planned / run)	100 / 31 (early stop, patience 15)
Best epoch	16 (lowest val loss)

E. Dataset

The UWF subset is split 80/20 with `random_state = 42` (random, not stratified): 8,323 training and 2,081 validation images, the latter used for evaluation.

F. Preprocessing

Each image is resized to 512×512 before the elliptical `apply_uwf_mask` (tightness = 0.98) is applied—resizing first keeps the heavy OpenCV masking from bottlenecking the data loader—and normalised with ImageNet statistics. Training augmentation uses random horizontal/vertical flips, colour jitter, and random rotation ($\pm 10^\circ$); validation applies only the resize, mask, and normalisation.

G. Architecture

The network is UWF_MultiScaleDRNet: a ResNet-50 backbone with four 1×1 lateral convolutions and a top-down, bilinearly-upsampled feature pyramid [6], a 7×7 spatial-attention gate [7] on the fused map, global average pooling, a 50% dropout layer, and two linear heads for the seven lesions and five grades. Relative to Experiment III the input is 512×512 (rather than 1024×1024) and an explicit dropout layer is added for regularisation.

H. Training Procedure

Training was scheduled for up to 100 epochs with early stopping (patience 15) on validation loss and a ReduceLROnPlateau schedule (factor 0.5, patience 3) that stepped the learning rate down at epochs 15, 21, 25, and 29. Training stopped at epoch 31, with the best (lowest-loss) checkpoint at **epoch 16**.

I. Hyperparameters

Table X lists the configuration.

J. Evaluation Metrics

Each epoch records grade accuracy, quadratic-weighted Cohen’s κ , and macro-averaged lesion F1 and AUC on the validation set.

TABLE XI
EXPERIMENT V — VALIDATION RESULTS

Metric	Epoch 16 (selected)	Best (epoch)
Grade accuracy	0.7578	0.7770 (22)
Grade κ (quad.)	0.9208	0.9313 (25)
Lesion macro-F1	0.7151	0.7311 (31)
Lesion macro-AUC	0.9504	0.9507 (26)

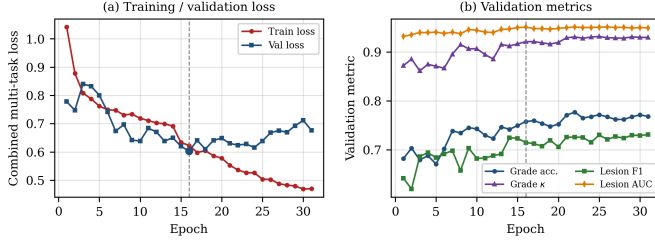


Fig. 10. Experiment V training dynamics over 31 epochs. (a) Training loss falls steadily while validation loss bottoms out at epoch 16 (dashed line) and then diverges—classic overfitting captured by the checkpoint policy. (b) Validation grade accuracy and quadratic κ climb into the high-0.9 range for κ , while lesion AUC holds near 0.95; the asymmetric loss yields strong ordinal grade agreement and stable lesion ranking.

K. Results

At the best-validation-loss checkpoint (epoch 16) the model attains grade accuracy **0.7578**, quadratic κ **0.9208**, lesion macro-F1 0.7151, and lesion AUC 0.9504. Grade accuracy continues to rise afterward—peaking at 0.7770 (epoch 22) with $\kappa = 0.9304$ —but the validation loss is already increasing, so the checkpoint policy retains epoch 16. Lesion AUC is stable near 0.95 throughout and lesion F1 peaks at 0.7311 (epoch 31). Table XI lists the selected-checkpoint and peak values, and Fig. 10 shows the trajectories.

L. Analysis

The asymmetric objective produces a strong ordinal grader—quadratic κ around 0.92 means most grade errors fall on adjacent severities—and a well-ranked lesion detector (AUC $\approx$ 0.95). The divergence between training and validation loss after epoch 16 shows the model begins to overfit the small, unstratified validation split, which the early-stopping-on-loss policy handles correctly.

M. Discussion

Compared with the frequency-and-risk weighting of Experiment IV, the asymmetric loss encodes the miss/false-alarm asymmetry more directly and yields a higher overall lesion macro-F1 (0.715 vs. 0.56) because it does not suppress the common lesions as aggressively. The lower input resolution (512 vs. 1024 in Experiment III) trades peripheral detail for a larger effective batch and faster epochs while retaining strong grade agreement.

N. Limitations

The model is selected on validation loss and the validation partition doubles as the evaluation set; the split is random and unstratified and no seed is fixed. The best checkpoint

was not retained on disk. The weakly-supervised bounding box is illustrative and unvalidated against ground-truth lesion locations.

O. Conclusion

An asymmetric, false-negative-penalised focal loss trains the UWF feature-pyramid network to a quadratic κ of 0.92 and lesion AUC of 0.95 at a grade accuracy of 75.8%, and its Grad-CAM maps support weakly-supervised lesion localization without bounding-box labels.

VIII. EXPERIMENT VI: SUPERVISED CONTRASTIVE LEARNING FOR RARE LESIONS (UWF)

A. Objective

Experiment VI attacks the rare-lesion problem in the *representation* rather than only the loss weighting, adding a supervised contrastive objective [8] on intermediate features so that images containing PDR lesions are pulled together and separated from those without.

B. Motivation

Loss re-weighting (Experiments IV–V) adjusts the decision boundary but leaves the feature space unchanged. If the embeddings of rare-lesion images are not already well separated, the classifier must work against an entangled representation. Supervised contrastive learning shapes the feature space directly, encouraging a cluster for rare-lesion images that the linear heads can then read out more reliably.

C. Methodology

A binary label—whether an image contains any of NV, VH, or RD—supervises a contrastive loss applied at two depths: an “early” projection of the `layer2` features and a “deep” projection of the `layer4` features, each mapped by a two-layer MLP to a 128-d embedding. The total objective is

$$L = L_{\text{asym}} + L_{\text{CE}} + 0.5 L_{\text{SupCon}}^{\text{early}} + 0.5 L_{\text{SupCon}}^{\text{deep}}, \quad (3)$$

where L_{asym} is the asymmetric focal lesion loss of Experiment V, L_{CE} the grade cross-entropy, and each SupCon term uses temperature $\tau = 0.1$. Classification still flows through the FPN-fused map; the contrastive heads act only as an auxiliary representational signal.

D. Experimental Setup

Implemented in PyTorch and trained across two CUDA GPUs with `DataParallel`. The same accuracy, quadratic κ , macro-F1, and AUC metrics as Experiment V are recorded each epoch.

E. Dataset

The UWF subset is split 80/20 with `random_state = 42` (random, not stratified): 8,323 training and 2,081 validation images. The contrastive label is derived per image as $\mathbb{I}[\text{NV} \vee \text{VH} \vee \text{RD}]$.

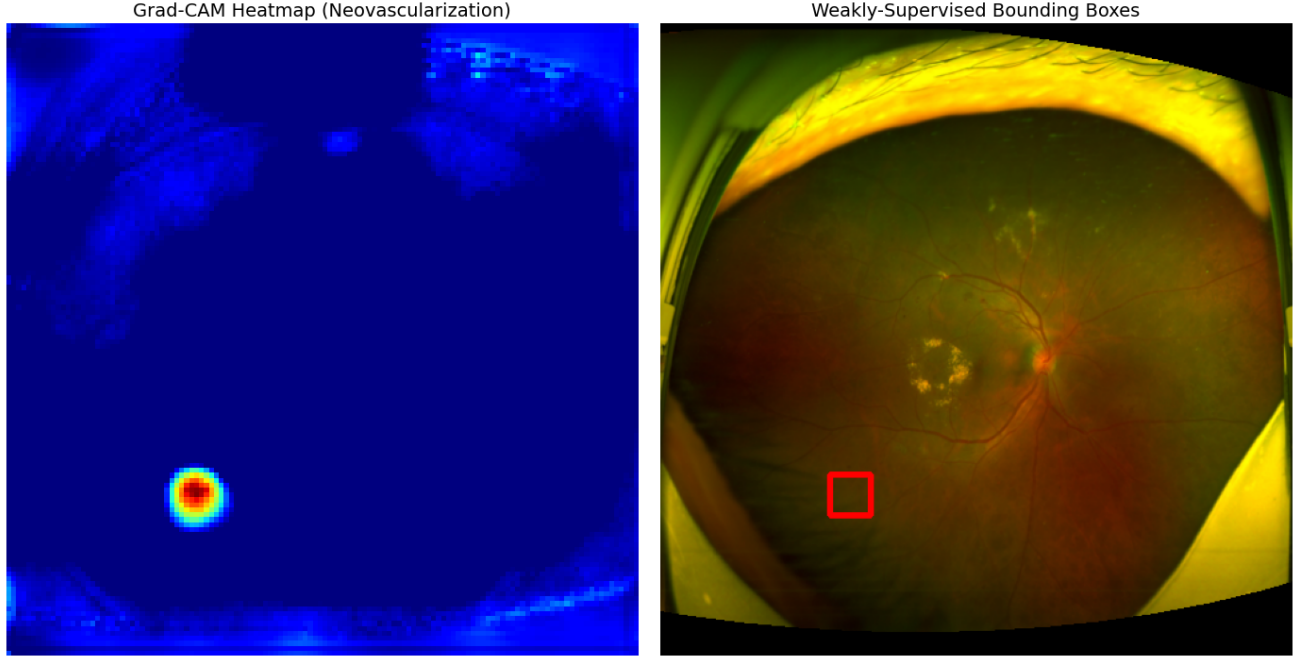


Fig. 11. Weakly-supervised localization for `UWF_MultiScaleDRNet`. Because the dataset has no bounding-box annotations, a Grad-CAM [4] heatmap for the neovascularization (NV) logit (left) is thresholded to derive a bounding box (right). The activation concentrates on a focal region of the masked retina, illustrating how the FN-penalised model can be used for approximate lesion localization without box labels.

F. Preprocessing

Identical to Experiment V: resize to 512×512 , elliptical `apply_uwf_mask` (`tightness` = 0.98), ImageNet normalisation, and flip/colour-jitter/rotation ($\pm 10^\circ$) augmentation on the training split only.

G. Architecture

The network `UWF_SupConDRNet` shares the ResNet-50 backbone and top-down feature pyramid [6] with Experiment V (without the spatial-attention gate) and adds two projection heads: `proj_early` ($512 \rightarrow 512 \rightarrow 128$) on globally-pooled `layer2` features and `proj_deep` ($2048 \rightarrow 512 \rightarrow 128$) on `layer4` features. The FPN-fused map is globally pooled and read out by linear lesion ($512 \rightarrow 7$) and grade ($512 \rightarrow 5$) heads.

H. Training Procedure

Training was scheduled for 60 epochs with early stopping (patience 10) on validation loss and `ReduceLROnPlateau` (factor 0.5, patience 3), with gradient-norm clipping at 2.0. The captured run is a *resumption* from a checkpoint at epoch 27; it continued through epoch 37 and then early-stopped because the validation loss never improved on the epoch-27 value—that is, the model had already converged by epoch 27. Consequently the best-loss checkpoint (the only preserved checkpoint on disk, `best_model_for_3.pth`) predates the captured log, and the metrics reported below characterise the converged plateau of epochs 28–37.

I. Hyperparameters

Table XII lists the configuration.

TABLE XII
EXPERIMENT VI — TRAINING CONFIGURATION

Setting	Value
Backbone	ResNet-50 + FPN + two SupCon heads
Compute	$2 \times$ GPU (DataParallel)
Input resolution	512×512
Optimiser	AdamW
Learning rate	1×10^{-4} (ReduceLROnPlateau)
Weight decay	1×10^{-4}
Batch size	8
Gradient clipping	max-norm 2.0
SupCon temperature	$\tau = 0.1$
Loss	Asym. focal + CE + 0.5 SupCon _{early} + 0.5 SupCon _{deep}
Epochs (planned)	60 (early stop, patience 10)
Captured run	Resumed epochs 28–37

J. Evaluation Metrics

Grade accuracy, quadratic-weighted Cohen’s κ , and macro-averaged lesion F1 and AUC, evaluated on the validation set each epoch.

K. Results

Over the converged plateau (epochs 28–37) validation grade accuracy holds between 0.7544 and **0.7645** (peak at epoch 31), quadratic κ between 0.9157 and 0.9223, lesion macro-F1 between 0.7243 and 0.7390, and lesion macro-AUC between 0.9501 and 0.9523—the highest lesion AUC among the UWF experiments. Training loss keeps falling while validation loss drifts slightly upward (Fig. 12), consistent with a converged model that is beginning to overfit. Table XIII reports the plateau ranges. Fig. 13 shows Grad-CAM localization on a genuine rare-lesion case.

TABLE XIII
EXPERIMENT VI — VALIDATION RESULTS (CONVERGED PLATEAU,
EPOCHS 28–37)

Metric	Plateau range	Peak (epoch)
Grade accuracy	0.754–0.765	0.7645 (31)
Grade κ (quad.)	0.916–0.922	0.9223 (36)
Lesion macro-F1	0.724–0.739	0.7390 (29)
Lesion macro-AUC	0.950–0.952	0.9523 (28)
Best-loss checkpoint predates the captured log (epoch ≤ 27).		

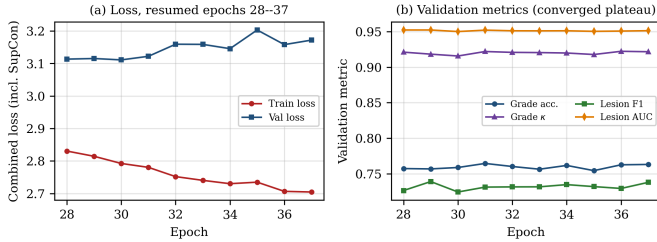


Fig. 12. Experiment VI training dynamics over the captured, resumed epochs 28–37. (a) Training loss (which includes the two contrastive terms and is thus not comparable in absolute value to the other experiments) continues to fall while validation loss drifts upward. (b) Validation grade accuracy, quadratic κ , lesion F1, and AUC are essentially flat, confirming that the model had already converged before the captured window.

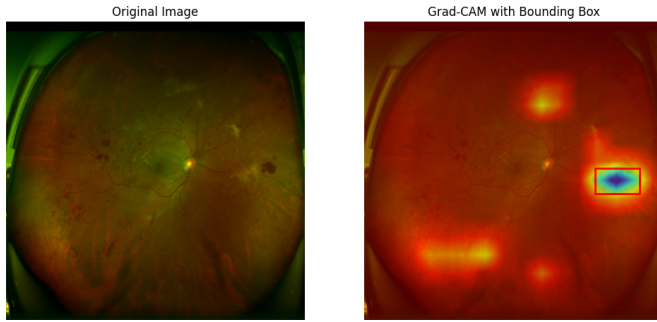


Fig. 13. Grad-CAM [4] localization for UWF_SupConDRNet on a validation image that genuinely contains a rare (NV) lesion. The heatmap is computed at `layer4` for the NV logit and overlaid on the masked UWF image (right), with a thresholded bounding box marking the peak activation. The contrastive auxiliary yields focal activation on a plausible peripheral region.

L. Analysis

On the converged plateau the contrastive model matches the asymmetric-loss model on grade accuracy and κ and attains the highest lesion AUC of the UWF group (≈ 0.952), consistent with the contrastive objective producing a better-ranked feature space for rare-lesion images. Because the captured metrics sit past the best-loss checkpoint, they represent a slightly overfit state and should be read as a lower bound on the selected model’s validation quality.

M. Discussion

The auxiliary contrastive signal (weight 0.5 at two depths) improves lesion ranking without a spatial-attention module, suggesting the representational and attention-based routes to rare-lesion sensitivity are partly interchangeable. The gains over the asymmetric-loss baseline are modest on grade metrics,

so the contrastive term is best viewed as a complementary regulariser rather than a standalone fix. This is also the only UWF rare-lesion experiment whose best checkpoint was preserved, making it the natural artifact for downstream reuse.

N. Limitations

The captured log begins at epoch 28 (a resumed session), so the exact best-epoch validation metrics are not recorded and the plateau values are reported instead. The combined loss is not comparable across experiments because of the additional contrastive terms. As elsewhere, the split is random and unstratified, validation doubles as evaluation, and no seed is fixed. The instantiated SupCon temperature ($\tau = 0.1$) differs from the class default (0.07).

O. Conclusion

Adding a two-depth supervised contrastive objective to the UWF multi-task network produces the highest lesion AUC of the UWF experiments (≈ 0.952) at a grade accuracy near 76% and quadratic κ of 0.92, indicating that shaping the feature space is a viable complement to loss re-weighting for rare-lesion sensitivity.

IX. EXPERIMENT VII: LEARNED UNCERTAINTY TASK-WEIGHTING (UWF)

A. Objective

Experiments IV–VI set the balance between the grade, lesion, and auxiliary objectives by hand. Experiment VII instead *learns* the multi-task weights during training, using the homoscedastic-uncertainty formulation of Kendall, Gal, and Cipolla [10], on top of the contrastive network of Experiment VI.

B. Motivation

The combined objective of Experiment VI weights four terms (lesion, grade, and two contrastive losses) with fixed coefficients (1:1:0.5:0.5). These are guesses, and the optimal balance may shift during training. Homoscedastic uncertainty weighting treats each task’s weight as a function of a learnable observation-noise parameter, letting the network raise the weight of tasks it can model confidently and lower it for noisier ones—removing a manual hyperparameter and providing a per-epoch diagnostic of how the model apportions capacity.

C. Methodology

For each task i a log-variance $s_i = \log \sigma_i^2$ is a free parameter, and the task losses (all non-negative) are combined as

$$L = \sum_i \left[\frac{1}{2} e^{-s_i} L_i + \frac{1}{2} s_i \right], \quad w_i = \frac{1}{2} e^{-s_i}, \quad (4)$$

the practical log-variance form [10]: the $\frac{1}{2} s_i$ term penalises a task simply switching itself off. The per-task weights w_i are logged every epoch as the primary diagnostic. The log-variances form their own optimiser parameter group (higher learning rate, zero weight decay) and are clamped to $|s_i| \leq 3$ for numerical safety. The four weighted tasks are the lesion, grade, and the two SupCon auxiliaries of Experiment VI.

TABLE XIV
EXPERIMENT VII — TRAINING CONFIGURATION

Setting	Value
Backbone	UWF_SupConDRNet (as Exp. VI)
Compute	2× GPU (DataParallel), AMP
Input resolution	512 × 512
Optimiser (network)	AdamW, lr 1×10^{-4} , wd 1×10^{-4}
Optimiser (log-vars)	lr 1×10^{-3} , wd 0
Weighting	Homoscedastic uncertainty, Eq. (4)
Weighted tasks	lesion, grade, SupCon _{early/deep}
Log-var init / clamp	$s_i=0 / s_i \leq 3$
Batch size	8
Gradient clipping	max-norm 2.0
Seed	42 (fixed)
Epochs (planned / run)	60 / 30
Best epoch (persisted log)	21 (lowest val loss)

D. Experimental Setup

Implemented in PyTorch across two CUDA GPUs (DataParallel) with AMP; a global seed is fixed for reproducibility. Grade quality is tracked with accuracy and quadratic-weighted Cohen’s κ ; lesion quality with macro-F1, macro-AUROC, macro PR-AUC (more informative than AUROC under heavy negative dominance), and per-rare-class recall.

E. Dataset

The UWF subset is split 80/20 with a fixed seed of 42 (random, not stratified): 8,323 training and 2,081 validation images, the latter used for evaluation.

F. Preprocessing

Identical to Experiment VI: resize to 512×512 , elliptical apply_uwf_mask (tightness = 0.98), ImageNet normalisation, and flip/colour-jitter/rotation ($\pm 10^\circ$) training augmentation.

G. Architecture

The backbone is Experiment VI’s UWF_SupConDRNet (ResNet-50 + FPN + two SupCon projection heads), unchanged. The learnable log-variances live in a separate UncertaintyWeightedLoss module rather than inside the network, so they are never duplicated by DataParallel and stay on a single device.

H. Training Procedure

Training was scheduled for 60 epochs with early stopping (patience 10) on validation loss and ReduceLROnPlateau (factor 0.5, patience 3), gradient clipping at 2.0, and the log-variances clamped after every step. The run proceeded to 30 epochs; the persisted per-epoch log covers epochs 7–24 and the checkpoint-and-weight history extends to epoch 30. The lowest validation loss in the persisted log is at **epoch 21**.

I. Hyperparameters

Table XIV lists the configuration.

TABLE XV
EXPERIMENT VII — VALIDATION RESULTS

Metric	Epoch 21 (selected)	Best (epoch)
Grade accuracy	0.7554	0.7645 (23)
Grade κ (quad.)	0.9198	0.9211 (30)
Lesion macro-F1	0.7356	0.7779 (30)
Lesion macro-AUROC	0.9605	0.9605 (21)
Lesion macro PR-AUC	0.8274	0.8395 (30)
Final-state rare recall: NV 0.891, VH 0.947, RD 0.958.		

TABLE XVI
EXPERIMENT VII — GRADE CONFUSION MATRIX (VALIDATION, FINAL STATE)

True \ Pred	0	1	2	3	4
0	614	60	1	0	1
1	121	211	67	0	0
2	3	56	304	76	5
3	0	0	68	194	7
4	2	4	10	28	249
Overall grade accuracy = $1572/2081 = 0.7554$.					

J. Evaluation Metrics

Grade accuracy and quadratic κ ; lesion macro-F1, macro-AUROC, macro PR-AUC; per-rare-class recall; and a grade confusion matrix from the error-analysis pass.

K. Results

The learned weights are the headline finding (Fig. 14). The lesion log-variance drives immediately to its clamp ($s_{\text{lesion}} = -3$), pinning the lesion weight at its ceiling ($w_{\text{lesion}} \approx 10.0$) from epoch 4 onward, while the grade weight rises smoothly from 0.5 to ≈ 1.2 and the two contrastive weights settle near 0.2. In other words the mechanism did not freely balance the lesion task—it saturated it—so the run behaves like a fixed, strongly lesion-weighted objective rather than a fully adaptive one.

Despite (and partly because of) this, the run attains the strongest lesion metrics of the whole study: validation lesion macro-F1 up to 0.778, macro-AUROC up to 0.961, and macro PR-AUC up to 0.840, exceeding the contrastive baseline of Experiment VI (0.739/0.952). The final-state error analysis shows excellent rare-lesion recall—NV 0.891 (26 false negatives of 238), VH 0.947 (12/226), RD 0.958 (2/48). The grade confusion matrix (Table XVI) is strongly near-diagonal, consistent with the quadratic $\kappa \approx 0.92$. Table XV lists the selected-checkpoint and peak values; Fig. 15 shows the trajectories.

L. Analysis

The clamp saturation means Experiment VII is best read as an empirical upper bound on what a heavily lesion-weighted version of the Experiment-VI network achieves: the strongest lesion F1/AUROC/PR-AUC and rare-class recall in the study. The grade weight’s smooth rise shows the uncertainty parameters *did* adapt for the well-behaved grade and contrastive tasks; only the lesion task hit the numerical boundary.

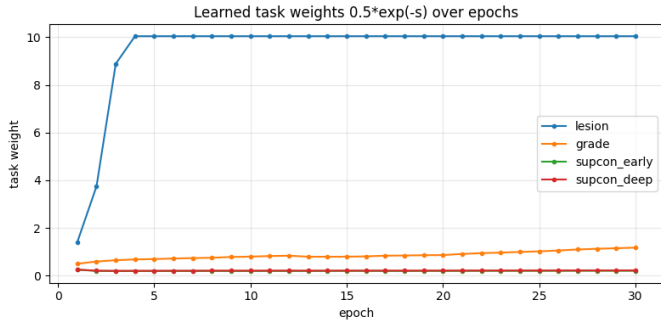


Fig. 14. Learned task weights $w_i = \frac{1}{2}e^{-s_i}$ over 30 epochs (Eq. (4)). The lesion weight saturates at its clamp ceiling (≈ 10.0) by epoch 4 and stays pinned, while the grade weight rises gradually to ≈ 1.2 and the two contrastive weights remain near 0.2. The saturation is the core diagnostic: the uncertainty mechanism strongly—but not adaptively—prioritised the lesion task.

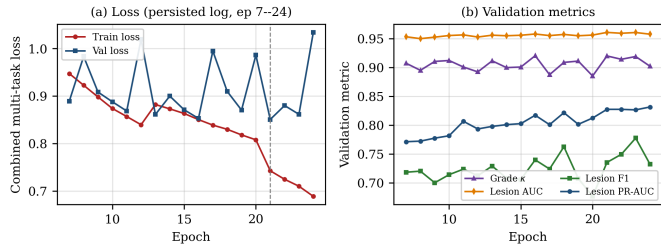


Fig. 15. Experiment VII training dynamics (persisted log, epochs 7–24). (a) Training loss falls steadily while validation loss oscillates around the epoch-21 minimum (dashed line). (b) Lesion macro-AUROC (≈ 0.96), grade κ (≈ 0.91), and lesion PR-AUC rise gently; PR-AUC climbs from 0.77 to 0.83, indicating improving ranking of the rare lesions.

M. Discussion

The result is a genuine improvement in rare-lesion detection but is only partly attributable to “learned” weighting—the lesion component effectively reverted to a large fixed weight. A wider clamp, a smaller log-variance learning rate, or the analytic softmax-normalised variant would let the lesion weight settle rather than rail, and would make the adaptivity claim cleaner. The two persisted checkpoints (`best_model (1).pth`, `checkpoint (2).pth`) preserve this state for reuse.

N. Limitations

The lesion log-variance is pinned at the clamp for the entire run, so the weighting is not fully adaptive. The persisted CSV covers epochs 7–24 while the run reached 30; peak values are drawn from the union of the CSV and the checkpoint log. The split is random and unstratified, validation doubles as evaluation, and although a seed is fixed the run is single-seed. The run used the four-task (“weight all tasks”) configuration rather than the grade-plus-lesion-only variant.

O. Conclusion

Homoscedastic-uncertainty weighting applied to the contrastive UWF network yields the study’s best rare-lesion detection (lesion F1 up to 0.778, AUROC up to 0.961; rare recall NV/VH/RD = 0.89/0.95/0.96), but the lesion

weight saturated at its clamp, so the gain reflects strong fixed prioritisation more than fully learned balancing.

X. EXPERIMENT VIII: RECALL-DRIVEN CURRICULUM RESAMPLING (UWF)

A. Objective

Experiment VIII addresses rare-lesion sensitivity from the *data-sampling* side. Holding the Experiment-VI network and loss fixed, it makes the training sampler the only independent variable: a recall-driven curriculum resampler (RDCR) that oversamples classes the model is currently missing.

B. Motivation

Loss re-weighting (Experiments IV–VII) changes how much each example counts; resampling changes which examples enter the batch. A static rarity-based sampler must guess a fixed oversampling ratio in advance. RDCR instead adapts the ratio online from validation recall, so effort tracks the classes that are actually being missed, and it fixes a standing weakness of the earlier experiments by using a *stratified* split.

C. Methodology

Twelve rebalancing channels (five grades, seven lesions) each carry an importance $I_k = \pi_k^a (D_k + \epsilon)^b$, where π_k is a static effective-number rarity prior [11] and D_k is an exponential-moving-average of the class *deficit* $1 - \text{recall}_k$ measured on validation each epoch. A sample’s weight collapses its channel importances by a max reduction (avoiding double-counting co-occurring PDR lesions). A curriculum temperature $\tau(\text{epoch})$ holds sampling uniform for a warmup (5 epochs) then ramps the importances in over 15 epochs, and the final per-sample weights are clamped to a maximum ratio of 20:1 to limit rare-image repetition. The sampler feeds a `WeightedRandomSampler`; the loss (asymmetric focal + CE + two SupCon terms) is unchanged. A CPU unit test asserts the sampler up-weights a deliberately starved rare class.

D. Experimental Setup

Implemented in PyTorch across two CUDA GPUs (`DataParallel`) with AMP, a fixed seed, and deterministic cuDNN. The sampler’s deficit state is checkpointed so a resumed run continues the controller and curriculum.

E. Dataset

The UWF subset is split 80/20 with seed 42, *stratified on the rare-lesion indicator*—the only experiment here to do so—giving 8,323 training and 2,081 validation images. Because the validation split differs from the random split of Experiments III–VII, its metrics are not directly comparable to them.

F. Preprocessing

Identical to Experiment VI: resize to 512×512 , elliptical `apply_uwf_mask` (`tightness` = 0.98), ImageNet normalisation, and `flip/colour-jitter/rotation` ($\pm 10^\circ$) training augmentation—augmentation is also the primary defence against oversampling overfitting, so repeatedly drawn rare images look different on each draw.

TABLE XVII
EXPERIMENT VIII — TRAINING AND SAMPLER CONFIGURATION

Setting	Value
Backbone / loss	UWF_SupConDRNet / folder-VI combined
Compute	2× GPU (DataParallel), AMP
Input resolution	512 × 512
Split	80/20, seed 42, stratified on rare flag
Optimiser	AdamW, lr 1×10^{-4} , wd 1×10^{-4}
Batch size	8
Gradient clipping	max-norm 2.0
Rarity prior	Effective-number [11], $\beta=0.999$, $a=0.5$
Deficit EMA	$m=0.7$, exponent $b=1$, floor $\epsilon=10^{-3}$
Curriculum	warmup 5 ep, ramp 15 ep ($\tau:0 \rightarrow 1$)
Weight-ratio clamp	$\rho_{\max} = 20$
Epochs (planned / logged)	60 / 18

G. Architecture

Experiment VI’s UWF_SupConDRNet (ResNet-50 + FPN + two SupCon heads), unchanged, so that any change in behaviour is attributable to the sampler alone.

H. Training Procedure

Training was scheduled for 60 epochs with early stopping (patience 10) on validation loss and ReduceLROnPlateau (factor 0.5, patience 3), gradient clipping at 2.0. Each epoch rebuilds the sampler from the previous epoch’s recall deficit and the current curriculum τ ; the warmup ends at epoch 5 and the ramp completes by epoch 20. The persisted metrics log covers 18 epochs.

I. Hyperparameters

Table XVII lists the configuration.

J. Evaluation Metrics

Quadratic-weighted Cohen’s κ (grade), lesion macro-F1 and macro-AUROC, and per-rare-class recall (NV/VH/RD), logged each epoch. Grade accuracy is not recorded in the persisted log; κ is the grade headline.

K. Results

As the curriculum engages after the epoch-5 warmup, grade κ and lesion AUROC rise together (Fig. 16a): κ climbs from ≈ 0.87 to a peak of **0.9234** (epoch 14), lesion AUROC to 0.9455 (epoch 18), and lesion macro-F1 to 0.7088 (epoch 17). The rare-class recalls (Fig. 16b) are the headline: RD recall reaches 1.0 at epochs 7–8 as the ramp begins, and NV/VH range over roughly 0.83–0.96, though all three are volatile—expected given only ~ 45 RD positives in the validation split. The lowest validation loss is at epoch 12 (2.369). Table XVIII summarises the values.

L. Analysis

The curriculum has the intended effect: rare-class recall improves as τ ramps, and grade κ reaches its peak (0.9234) in the same window, so rebalancing the sampler did not harm grading. The volatility of RD recall is a direct consequence of the tiny rare-positive count and is why per-epoch recall, not a single final number, is the honest way to report it.

TABLE XVIII
EXPERIMENT VIII — VALIDATION RESULTS (STRATIFIED SPLIT)

Metric	Epoch 12 (selected)	Best (epoch)
Grade κ (quad.)	0.9144	0.9234 (14)
Lesion macro-AUROC	0.9423	0.9455 (18)
Lesion macro-F1	0.7011	0.7088 (17)
RD recall	0.889	1.000 (7–8)
NV recall	0.879	0.957 (1)
VH recall	0.914	0.951 (1)
Stratified split—not directly comparable to Exp. III–VII.		

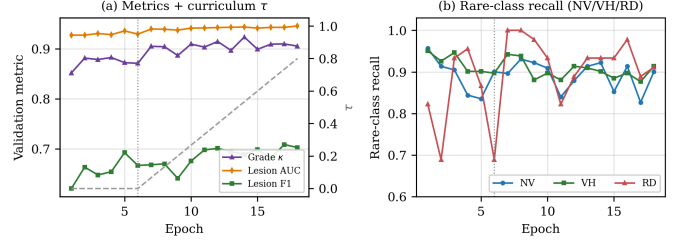


Fig. 16. Experiment VIII dynamics over 18 logged epochs. (a) Grade κ , lesion AUROC, and lesion F1 rise as the curriculum τ (dashed, right axis) ramps in after the epoch-5 warmup (dotted line). (b) Per-rare-class recall: RD reaches 1.0 once the ramp engages; NV/VH remain high but volatile, reflecting the very small number of rare positives in the validation split.

M. Discussion

Because the backbone and loss are frozen, the gains isolate the sampler’s contribution. On its own (stratified) validation set RDCR matches the mature folder-VI baseline on grade κ (≈ 0.92) and lesion AUROC (≈ 0.945) while pushing rare recall high—but it does not clearly beat the baseline, and the comparison is confounded by the different split. The stratified split is itself the more rigorous protocol and is the main methodological contribution here.

N. Limitations

Only 18 of 60 planned epochs are logged, and no checkpoint was preserved on disk. The run is single-seed. Grade accuracy is absent from the log. Most importantly, the stratified split makes the numbers non-comparable to the other experiments, so “matched baseline” is a qualitative statement, not a controlled head-to-head; a like-for-like comparison would require rerunning the baseline on the same stratified split with multiple seeds.

O. Conclusion

Recall-driven curriculum resampling, with the network and loss held fixed, lifts rare-class recall (RD to 1.0, NV/VH ≈ 0.9) and reaches grade $\kappa = 0.923$ and lesion AUROC 0.946 on a stratified split, demonstrating adaptive sampling as a viable, backbone-agnostic route to rare-lesion sensitivity, pending a multi-seed, same-split comparison.

XI. EXPERIMENT IX: SHARED VS. SEPARATE LESION HEADS (UWF)

A. Objective

Experiment IX is a controlled architectural ablation: does giving the rare PDR lesions their own *dedicated* classification head improve their detection, once the effect is separated from the effect of simply adding head capacity? It holds the entire trunk and loss fixed and varies only the lesion head, on a proper held-out test split.

B. Motivation

In a single shared lesion head, the gradient from the 6–30× more frequent common lesions can dominate the parameters that also serve the rare lesions. A natural hypothesis is that a *separate* rare-lesion head, decoupled from the common lesions, would be freed from this domination and detect NV/VH/RD better. But a separate head also adds parameters, so a fair test needs a capacity-matched control that keeps the head shared—otherwise any gain could be mere capacity.

C. Methodology

Three head variants share an identical trunk and differ only in `classifier_lesions`: **A (shared)**, a single `Linear(512, 7)` (3,591 head params); **A' (shared-MLP)**, a two-layer MLP over all seven lesions (66,567 params); and **B (separate)**, a linear head for the four common lesions plus a dedicated MLP head for the three rare lesions (68,103 params). A' and B are capacity-matched, so the contrast $B - A'$ isolates *separation* while $A' - A$ isolates *capacity*. The split head concatenates `[common | rare]` logits, which is exactly the canonical lesion order, so the loss and metrics are variant-agnostic. Six CPU unit tests verify the design: identical trunk size across variants (26,921,797 params), exact head-parameter counts, rare-logit gradients touching only the rare sub-head, loss reduction-consistency (split = shared), and that per-lesion threshold tuning beats a fixed 0.5 on rare recall.

D. Experimental Setup

Implemented in PyTorch across two CUDA GPUs (`DataParallel`) with AMP, fixed seeds, and deterministic `cuDNN`. Model selection uses a composite score ($0.5 \times \text{rare recall} + 0.5 \times \text{macro-F1}$) on validation, and per-lesion decision thresholds are tuned on validation before the test evaluation. Unusually for this study, evaluation uses a genuine held-out test split, not the validation set.

E. Dataset

The UWF subset is divided into a *three-way* 70/15/15 train/validation/test split, stratified on the rare-lesion indicator with seed 42: 7,282 training, 1,561 validation, and 1,561 test images, with a matched rare-positive fraction of ≈ 0.149 in every partition. All reported numbers are on the held-out test split.

F. Preprocessing

The standard UWF pipeline: resize to 512×512 , elliptical `apply_uwf_mask` (`tightness` = 0.98), ImageNet normalisation, and flip/colour-jitter/rotation training augmentation.

TABLE XIX
EXPERIMENT IX — TRAINING CONFIGURATION

Setting	Value
Trunk	UWF_SupConDRNet (ResNet-50 + FPN)
Variable component	Lesion head only (A / A' / B)
Head params (A/A'/B)	3,591 / 66,567 / 68,103
Compute	2× GPU (<code>DataParallel</code>), AMP
Input resolution	512 × 512
Split	70/15/15, seed 42, stratified (held-out test)
Optimiser	AdamW, lr 1×10^{-4} , wd 1×10^{-4}
Batch size	16
Loss	Asym. focal + CE (LS 0.05) + 0.5 SupCon×2
Selection	Composite (0.5 rare recall + 0.5 macro-F1)
Thresholds	Per-lesion, tuned on validation
Epochs (this pass / full)	20 × 1 seed / 60 × 3 seeds

G. Architecture

MultiHeadDRNet is the folder-VI UWF_SupConDRNet trunk (ResNet-50 + FPN + two SupCon projection heads + a grade head), byte-identical across variants, with only the lesion head swapped. The shared trunk, grade head, and contrastive heads are therefore controlled, so any measured difference is attributable to the lesion head alone.

H. Training Procedure

The combined loss is the folder-VI objective (asymmetric focal on lesions, grade cross-entropy with label smoothing 0.05, and two SupCon terms). The reported run is a single-seed QUICKPASS of 20 epochs per variant (the full protocol is 60 epochs × 3 seeds, designed and resumable but not run here). Because the run was assembled across resumable sessions, the shared and shared-MLP variants were continued from earlier checkpoints while the separate variant trained afresh and early-stopped at epoch 14; the training budgets are therefore not perfectly matched, so the comparison is preliminary.

I. Hyperparameters

Table XIX lists the configuration.

J. Evaluation Metrics

On the held-out test split: lesion macro-F1, mean rare-class (NV/VH/RD) recall, and grade quadratic-weighted Cohen's κ , computed with the tuned per-lesion thresholds.

K. Results

Table XX and Fig. 17 give the held-out test results. The dedicated rare head (B) does *not* help: its mean rare-class recall (0.816) is the lowest of the three and well below the capacity-matched shared-MLP control (A', 0.890) and even the minimal shared linear head (A, 0.884). Reading the two contrasts, capacity alone ($A' - A$) moves rare recall by only +0.006 and slightly lowers macro-F1 (0.698 vs. 0.707), while separation ($B - A'$) *reduces* rare recall by 0.074. The plain shared head attains the best lesion macro-F1 (0.707) and a grade κ (0.914) on par with A' (0.915).

TABLE XX
EXPERIMENT IX — HELD-OUT TEST RESULTS (SINGLE-SEED,
20-EPOCH PASS)

Variant	Head params	Rare rec.	Macro-F1	κ
A (shared linear)	3,591	0.884	0.707	0.914
A' (shared MLP)	66,567	0.890	0.698	0.915
B (separate)	68,103	0.816	0.689	0.886

Rare rec. = mean recall over NV/VH/RD. Best per column in **bold**.

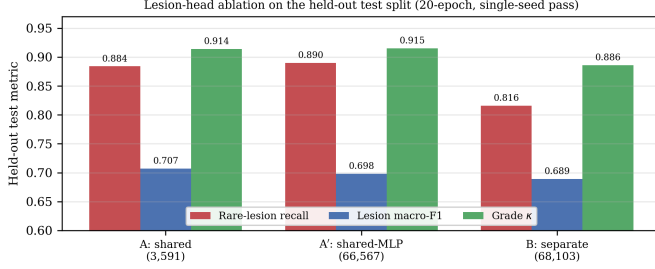


Fig. 17. Held-out test metrics for the three capacity-controlled lesion heads. Adding head capacity (A→A') barely changes any metric, and separating the rare head (A'→B) *lowers* rare-lesion recall from 0.890 to 0.816. The simple shared linear head (A) is the strongest on macro-F1 and competitive on rare recall and κ .

L. Analysis

The controlled design turns a plausible intuition into a testable claim and refutes it: on this preliminary single-seed pass, neither extra head capacity nor an explicit rare/common split improves held-out rare-lesion detection over a plain Linear(512, 7) head. The likely reason is that the rare-lesion signal is limited by the trunk features and the rare-positive count, not by head-level gradient competition, so a bigger or decoupled head mostly adds variance (and B's fresh, shorter training compounds this).

M. Discussion

The value here is methodological: a capacity-matched control (A') and a genuine held-out test split make the negative result trustworthy, and the unit tests guarantee the variants differ only where intended. It suggests that, for this backbone, rare-lesion gains are more likely to come from the trunk, the loss, or the data (Experiments IV–VIII) than from the head topology. The tuned-threshold step is a cheap, orthogonal lever that helped rare recall in the unit test and is retained for all variants.

N. Limitations

This is a single-seed, 20-epoch preliminary pass, not the pre-registered 60-epoch $\times 3$ -seed protocol, and the variants were trained across sessions with unequal effective budgets (B trained fresh and early-stopped while A/A' resumed), so the ranking may shift under matched, multi-seed training with a significance test. No checkpoints were retained locally.

O. Conclusion

A controlled, capacity-matched ablation on a held-out test split finds that a dedicated rare-lesion head does not improve—and in this pass slightly harms—rare-lesion detection relative

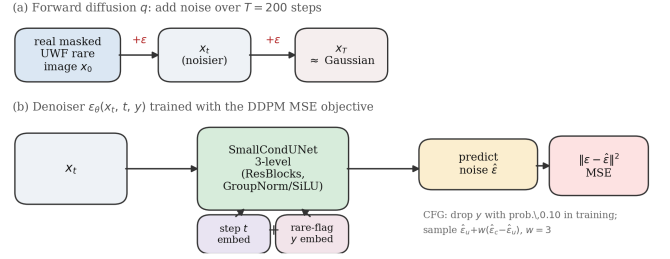


Fig. 18. Conditional-DDPM pipeline (method schematic drawn from the notebook code, not an experimental result). (a) The forward process noises a real masked rare-lesion retina over $T=200$ steps. (b) SmallCondUNet $\epsilon_\theta(x_t, t, y)$ predicts the added noise under an MSE objective, conditioned on the timestep and rare-flag embeddings; classifier-free guidance drops the condition with probability 0.10 in training and extrapolates with scale $w=3$ at sampling.

to a shared head (0.816 vs. 0.890 mean rare recall), indicating that head topology is not the bottleneck for rare-lesion sensitivity in this pipeline.

XII. EXPERIMENT X: SYNTHETIC MINORITY GENERATION (UWF)

A. Objective

Experiment X generates *new* synthetic rare-lesion images with a conditional diffusion model, as an alternative to over-sampling the same scarce positives, and runs the decisive test: does adding those images to training raise rare-class recall on a 100%-real held-out test set, versus a real-only baseline and versus classic oversampling? A full-scale generator was trained and the downstream study was run to completion; the result is a clean negative verdict.

B. Motivation

With only 238 retinal-detachment positives (2.3% of UWF), oversampling can only duplicate the same few images, so the classifier memorises them rather than learning what the lesion looks like. A generative model that synthesises plausible new rare-lesion retinas could add genuine variety, potentially improving rare-class recall beyond what duplication achieves.

C. Methodology

The runnable proof-of-concept is a compact, from-scratch *conditional pixel-space DDPM* [12]. The forward process adds Gaussian noise to a real masked retina over $T=200$ steps; a small U-Net $\epsilon_\theta(x_t, t, y)$ is trained to predict that noise by the DDPM simple objective $\mathbb{E}\|\epsilon - \epsilon_\theta\|^2$, conditioned on a rare-flag y . Classifier-free guidance [13] is enabled by dropping the condition with probability 0.10 during training and, at sampling, extrapolating $\hat{\epsilon} = \epsilon_\theta(x_t, \emptyset) + w(\epsilon_\theta(x_t, y) - \epsilon_\theta(x_t, \emptyset))$ with guidance scale $w=3$ (Fig. 18). The documented production generator is a LoRA-fine-tuned latent-diffusion model [14] at 256×256 ; the compact pixel-space slice shares the same objective at a size that trains within a session.

TABLE XXI

EXPERIMENT X — PROOF-OF-CONCEPT GENERATOR CONFIGURATION

Setting	Value
Generator	Conditional pixel-space DDPM (SmallCondUNet)
Resolution	64×64 (production target 256)
Diffusion steps T	200 (production ~ 1000)
Objective	DDPM ϵ -prediction MSE [12]
Guidance	Classifier-free [13], drop 0.10, $w=3$
Optimiser	AdamW, lr 2×10^{-4} , cosine
Batch size	16
Gradient clipping	max-norm 1.0
Generator params	3.86 M (SmallCondUNet)
Run	full, $2 \times T4$, ep 41/50 ($\approx 26.6k$ steps)
Best epoch-avg loss	0.0161 (epoch 39)
Production generator	LoRA latent diffusion [14] (documented)

TABLE XXII

EXPERIMENT X — DOWNSTREAM VERDICT ON THE 100%-REAL TEST SET (SEED 42)

Metric (real test)	real	+oversample	+synthetic
Mean rare recall	0.878	0.807	0.833
NV recall	0.894	0.806	0.888
VH recall	0.900	0.842	0.900
RD recall	0.839	0.774	0.710
Rare macro-F1	0.687	0.714	0.656
Rare macro PR-AUC	0.653	0.797	0.665
Grade κ (guardrail)	0.902	0.880	0.892
Common-lesion F1 (guardrail)	0.723	0.717	0.716
Best per row in bold . Verdict rule: synthetic must beat <i>both</i> other columns on mean rare recall without regressing the guardrails $\rightarrow$ not met .			

D. Experimental Setup

Implemented in PyTorch with AMP and optional multi-GPU, and structured with checkpoint/resume and dataset backup for multi-session training. Optional generative libraries are guarded so the pipeline runs on the pre-installed stack alone.

E. Dataset

The generator is trained on the UWF rare-positive images (any of NV/VH/RD present; 1,555 images, with RD only 238). Provenance is tracked so synthetic images are always labelled synthetic and never stored as real patient data (MMRDR is CC0).

F. Preprocessing

Each retina is resized to 512, masked with the elliptical `apply_uwf_mask`, downscaled to the generator resolution (64×64 in the slice), and normalised to $[-1, 1]$; a one-time cache of masked tensors speeds up epochs.

G. Architecture

SmallCondUNet is a three-level conditional U-Net of residual blocks (GroupNorm/SiLU). Conditioning enters by adding a sinusoidal timestep embedding and a learned rare-flag class embedding; a learned null token supports classifier-free guidance.

H. Training Procedure

AdamW (lr 2×10^{-4}) with a cosine schedule, AMP, gradient clipping at 1.0, and rolling/best checkpointing to a Kaggle dataset for multi-session resume. The reported run used `MODE=full` on two T4 GPUs over the real UWF rare-positive set (650 steps/epoch, ≈ 66 s/epoch), reaching epoch 41 of a planned 50; the best epoch-average loss was at epoch 39.

I. Hyperparameters

Table XXI lists the slice configuration.

J. Evaluation Metrics

Generative quality is checked with FID [15] and unbiased KID, computed between real and synthetic rare-lesion images at both the generator’s native 64×64 and the study’s 256×256 resolution, each against a real-vs-real floor. The decisive metric, however, is *downstream utility*: per-rare-class recall on a fixed, 100%-real held-out test set when the folder-VI classifier (UWF_SupConDRNet) is trained under three regimes—real only, real+oversampling, and real+synthetic—on one frozen stratified 70/15/15 split (train 7,282 / val 1,561 / test 1,561, signature `dalec8e21f22`). The validation and test sets are 100% real and identical across regimes; per-lesion thresholds are fitted on validation only. Grade quadratic κ and common-lesion macro-F1 act as collateral-damage guardrails.

K. Results

Generator. The DDPM ϵ -prediction MSE fell from 0.081 at epoch 1 to a best epoch-average of **0.0161** at epoch 39 (Fig. 19), converging stably; classifier-free sampling produces diverse conditional images (mean pairwise sample distance 29.9, i.e. no mode collapse).

Generation quality is poor at this scale. Synthetic-vs-real FID is 200.0 at 64px and 246.9 at 256px, against real-vs-real floors of only 26.8 and 29.2 (KID 0.256 and 0.326). The synthetic images are therefore far from the real distribution. Fig. 20 shows why: the samples reproduce the elliptical field of view and the green/orange UWF colouration but contain no recognisable vasculature or lesion structure.

The downstream verdict is negative. Table XXII and Fig. 21 give the held-out-test results. The real-only baseline is already strong on rare-class recall (0.878); adding synthetic images *lowers* it to 0.833, and classic oversampling lowers it further to 0.807. Synthetic augmentation also slightly regressed both guardrails (grade κ 0.892 vs. 0.902; common-lesion F1 0.716 vs. 0.723). By the pre-registered rule—beat *both* real and oversampling on mean rare recall without regressing the guardrails—the verdict is **False**: at proof-of-concept resolution, generated data does not improve rare-lesion detection. The damage is concentrated in retinal detachment, the rarest class (31 test positives), whose recall falls from 0.839 (real) to 0.710 (synthetic).

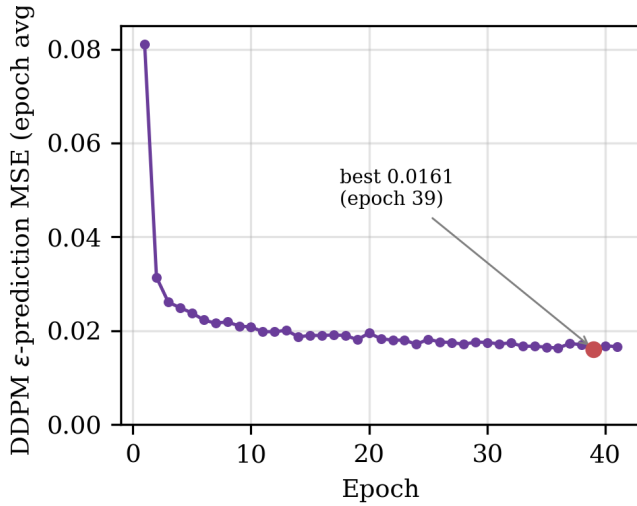


Fig. 19. Experiment X generator training. The DDPM noise-prediction MSE (epoch average) decreases from 0.081 to a best of 0.0161 at epoch 39 over the 41-epoch full run on the real UWF rare-positive set, indicating stable convergence of the conditional diffusion model.

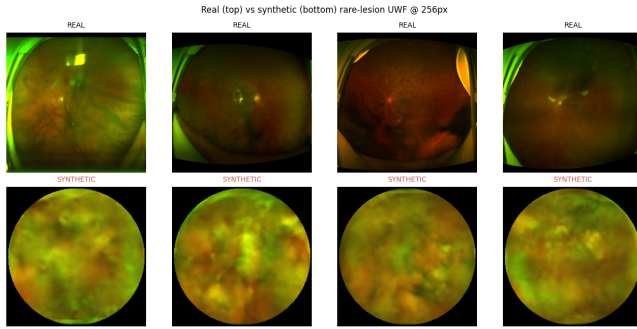


Fig. 20. Real UWF retinas (top) versus rare-conditioned synthetic samples (bottom, guidance $w=3$), shown at the study’s 256×256 . The generator captures the field of view and colouration but no vasculature or lesion structure, consistent with the very high synthetic-vs-real FID (246.9 vs. a 29.2 floor). Synthetic images are always labelled synthetic and never treated as real patient data.

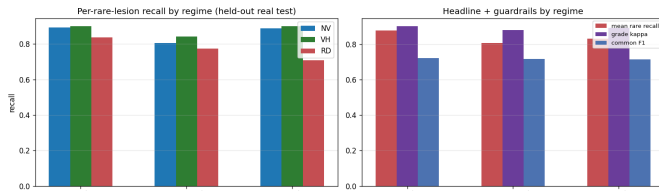


Fig. 21. Experiment X downstream verdict on the held-out real test set. Left: per-rare-lesion recall by training regime; adding synthetic data does not exceed the real-only baseline, and RD recall drops. Right: the headline (mean rare recall) and the two guardrails (grade κ , common-lesion F1) by regime; the real-only bars are highest or tied throughout.

L. Analysis

The result is internally consistent: the FID gate predicts it. Synthetic images 200–250 FID units from the real manifold cannot teach a classifier what a lesion looks like, so at best they add noise and at worst they mislead—exactly what the RD column shows. Two secondary observations refine the picture. First, the real-only baseline is unusually strong on rare recall

(0.878) because the asymmetric-focal + contrastive objective is already tuned for the rare classes, so there is little headroom for augmentation to add. Second, oversampling improves rare *ranking* (PR-AUC 0.797 vs. 0.653) while lowering thresholded recall, i.e. it trades sensitivity for precision—useful to note, but the clinically headline number, recall, still favours the plain real baseline.

M. Discussion

This is a clean negative result, and a useful one: at a proof-of-concept 64×64 resolution, generative augmentation beats neither the real-only baseline nor simple oversampling for rare-lesion detection. Because the whole pipeline is now built and measured on a 100%-real test set, the finding is trustworthy rather than anecdotal, and it localises the bottleneck precisely—generation *fidelity*, quantified by the FID gate, not the augmentation methodology. The clear next step is the higher-resolution latent-diffusion generator the design earmarks; only once its synthetic-vs-real FID approaches the real-vs-real floor would a downstream benefit be plausible.

N. Limitations

The downstream study is single-seed (QUICK_PASS), so no significance test accompanies it; the full protocol (≥ 3 seeds, synthetic amounts $\{0.5, 1, 2\} \times$, Wilcoxon) is implemented and gated for a later run. Synthetic images are $4\times$ upsampled from 64 to 256, a resolution-domain confound quantified by the 64-vs-256 FID gap. Each synthetic image inherits the full grade+lesion label of a random real rare-positive, which is weak supervision on the grade and common-lesion channels—one reason the guardrails are watched. The generator itself is the 64px pixel-space proof-of-concept, not the documented production LoRA latent-diffusion model.

O. Conclusion

With the generative-augmentation pipeline fully implemented and evaluated on a 100%-real held-out test set, synthetic rare-lesion images from the 64px DDPM do *not* improve rare-class recall over a real-only baseline (0.833 vs. 0.878) or over classic oversampling, and the very high synthetic-vs-real FID (≈ 247 at 256px) identifies generation fidelity as the reason. The negative verdict is itself the contribution: it establishes the evaluation and shows that a higher-resolution generator, not a different augmentation recipe, is what a future gain would require.

XIII. EXPERIMENT XI: CROSS-DATASET ZERO-SHOT AND FEW-SHOT GENERALIZATION

A. Objective

Every preceding experiment trains and evaluates on MM-RDR. Experiment XI asks the question that decides whether any of it generalises: taken off the shelf, how well do the trained models grade diabetic retinopathy on *unseen* datasets from different cameras and populations—first with no adaptation at all (*zero-shot*), then given only a handful of labelled examples per class (*few-shot*)? This is a grade-only study,

because the five-class ICDR severity scale (0–4) is the one label space shared across public DR datasets; the seven-lesion vocabulary is MMRDR-specific and does not transfer.

B. Motivation

In-domain accuracy can reward dataset-specific shortcuts (a camera’s colour cast, a site’s grading habits) that collapse on new data. A cross-dataset test separates a model that learned transferable retinal pathology from one that memorised a distribution. Foundation models such as RETFound [5] are pretrained precisely to transfer, so a like-for-like comparison against a task-specific CNN, on data neither has seen, is the fair way to judge that claim.

C. Methodology

Three trained models are frozen and applied to two external datasets. *Zero-shot* uses each model’s own grade head directly, taking $\arg \max$ over its five logits on the external test split—no fitting whatsoever. *Few-shot* freezes the backbone, extracts the penultimate feature once, and fits a fresh multinomial logistic head (standardised features) on k labelled images per class sampled from an external *support* pool, then predicts the held-out test split. This linear-probe design is the standard low-variance transfer probe: it isolates feature quality from the original head’s calibration. Each (k) setting is repeated over five independent support draws; we report the mean and standard deviation, with the zero-shot score as the $k=0$ anchor on the same curve.

D. Experimental Setup

Implemented in PyTorch on a dual-T4 Kaggle instance. Checkpoints are attached as read-only inputs and loaded with prefix-stripped state dicts (all three loaded with zero missing or unexpected tensors). Features are extracted in a single inference pass per split; the logistic probe is fitted on CPU via scikit-learn. The clean APTOS labels are auto-preferred when present; the harness detects and reports its inputs so the run is reproducible.

E. Datasets

Two genuinely unseen colour-fundus datasets, both on the same 0–4 ICDR scale: **APTOS 2019** [16] (the Kaggle Blindness-Detection competition data; 3,662 labelled images, Indian population) as the clean primary, split 50/50 stratified into support/test (1,831 each); and **IDRiD** [17] Disease-Grading, using its official train (1,239) / test (309) partition (an augmented mirror; see Limitations). Neither appears in any MMRDR training set, and both differ from MMRDR in camera and cohort, so this is a real domain shift.

F. Preprocessing

Each external image is preprocessed to match the corresponding model’s own training transform: resize to the model’s native size (512 for the CFP models, 256 for the UWF control), ImageNet normalisation, and—for the multi-scale CNN

TABLE XXIII
EXPERIMENT XI — ZERO-SHOT GRADE TRANSFER TO UNSEEN DATASETS

Data	Model	QWK	Acc	Bal. acc	Mac-F1
APTOS	MS-LAN (CFP)	0.854	0.705	0.528	0.505
APTOS	RETFound (CFP)	0.860	0.711	0.528	0.495
APTOS	UWF SupCon (ctrl)	0.689	0.578	0.482	0.412
IDRiD	MS-LAN (CFP)	0.626	0.511	0.481	0.478
IDRiD	RETFound (CFP)	0.593	0.476	0.398	0.404
IDRiD	UWF SupCon (ctrl)	0.598	0.375	0.431	0.364

Best per dataset in **bold**. Grade QWK = quadratic-weighted κ ;
Bal. acc = mean per-class recall. APTOS test $n=1,831$, IDRiD $n=309$.

only—the same circular fundus mask used in Experiment I. The UWF control receives standard CFP preprocessing (no ultra-widefield elliptical mask, whose geometry assumes a UWF field of view).

G. Models Under Test

Two CFP models and one deliberate negative control: the **multi-scale dual-branch CNN** (MS-LAN, Experiment I; grade head $\arg \max$; 512px); the **RETFound ViT-L/16** fine-tune (Experiment II; 512px; 1024-d pooled feature); and the **UWF supervised-contrastive network** (Experiment VI) as a *control* trained on the wrong modality (ultra-widefield), expected to transfer worst to standard fundus photographs.

H. Evaluation Metrics

The headline is the quadratic-weighted Cohen’s κ (QWK), the standard ordinal DR-grading metric, which is robust to the large shift in grade prior between datasets. Accuracy, balanced accuracy (mean per-class recall), and macro-F1 are reported alongside; a 5×5 confusion matrix is saved per model per dataset.

I. Results

Zero-shot (Table XXIII, Fig. 22). On the clean APTOS benchmark both CFP models transfer strongly and near-identically—RETFound QWK **0.860**, MS-LAN 0.854—with the UWF control clearly behind at 0.689. On the smaller IDRiD set all three compress into 0.59–0.63 (MS-LAN **0.626**), reflecting a harder shift and a 309-image test set. Throughout, QWK far exceeds raw accuracy (e.g. 0.86 vs. 0.71 on APTOS): the models get the ordinal severity right even when the exact class is off, so their errors sit near the confusion diagonal.

Few-shot (Table XXIV, Fig. 23). A tiny labelled probe recovers the domain gap where the source head is most mismatched. RETFound gains most on IDRiD (0.593 $\rightarrow$ 0.711 at $k=20$, +0.118), and the wrong-modality UWF control climbs steeply on APTOS (0.425 at $k=1$ to 0.759 at $k=20$)—its frozen features are usable even though its own head is not. On clean APTOS, however, the CFP models’ *own* fine-tuned heads already beat a k -shot linear probe (MS-LAN 0.854 $\rightarrow$ 0.776; RETFound essentially flat), a legitimate sign that little headroom remains when the source head is already well matched.

TABLE XXIV
EXPERIMENT XI — FEW-SHOT GRADE QWK VS. SHOTS PER CLASS
(MEAN $\pm$ STD, 5 EPISODES)

Model	$k=0$	$k=1$	$k=5$	$k=10$	$k=20$
<i>APTOS 2019 (clean)</i>					
MS-LAN	0.854	.691 $\pm$.03	.749 $\pm$.04	.770 $\pm$.01	.776 $\pm$.03
RETFound	0.860	.813 $\pm$.05	.844 $\pm$.01	.851$\pm$.00	.850 $\pm$.01
UWF (ctrl)	0.689	.425 $\pm$.15	.575 $\pm$.13	.712 $\pm$.02	.759 $\pm$.03
<i>IDRiD (augmented mirror)</i>					
MS-LAN	0.626	.540 $\pm$.09	.591 $\pm$.06	.627 $\pm$.04	.676 $\pm$.04
RETFound	0.593	.589 $\pm$.07	.660 $\pm$.05	.697 $\pm$.03	.711$\pm$.03
UWF (ctrl)	0.598	.254 $\pm$.12	.480 $\pm$.04	.552 $\pm$.03	.575 $\pm$.02

$k=0$ is the zero-shot anchor (each model's own head). $k>0$ is a frozen-backbone logistic probe on k images/class. IDRiD few-shot is optimistic (augmented mirror).

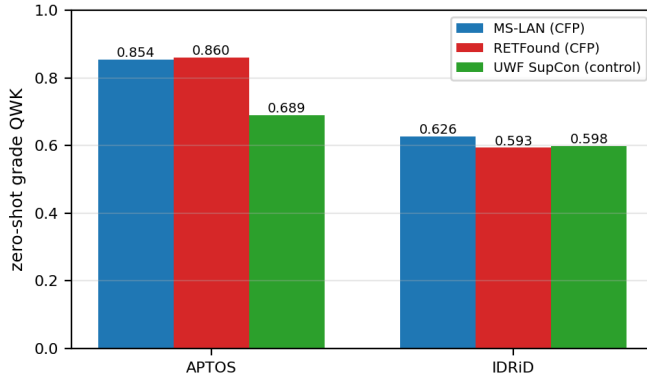


Fig. 22. Experiment XI zero-shot grade QWK on two unseen external datasets. On the clean APTOS benchmark the two CFP models transfer strongly and near-identically (RETFound 0.860, MS-LAN 0.854) while the ultra-widefield control lags at 0.689; on the harder, smaller IDRiD set the three compress to 0.59–0.63.

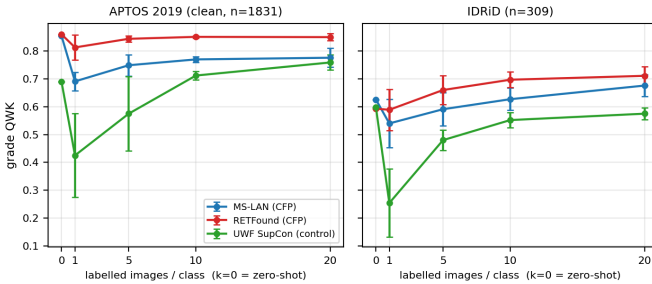


Fig. 23. Experiment XI few-shot transfer: grade QWK versus labelled images per class, with the zero-shot score as the $k=0$ anchor (mean $\pm$ std over five support draws). A small probe recovers the domain gap most where the source head is mismatched (RETFound on IDRiD; the wrong-modality UWF control on APTOS), whereas on clean APTOS the CFP models' own heads already match or exceed the probe.

J. Analysis

Three patterns are consistent across both datasets. First, *the foundation model is the best all-round generaliser*: RETFound is top zero-shot on the clean benchmark and shows the largest few-shot recovery on the shifted one, exactly the behaviour its pretraining promises. Second, *modality matters*: the UWF control transfers measurably worse than the CFP models on the clean benchmark (0.689 vs. ~ 0.86 QWK, and the lowest balanced accuracy everywhere). It does not collapse to chance—grade QWK is ordinal-tolerant, and a wrong-

modality retinal network still ranks severity partially—but the gap confirms the CFP models' transfer is modality-matched rather than a metric artefact. Third, *few-shot value is inversely related to source-head fit*: the probe helps precisely when the original head is domain-shifted and adds nothing when it is already well calibrated.

K. Discussion

The concluding message of the report is positive: the CFP models trained on MMRDR do not merely fit their own distribution—they grade an unseen dataset zero-shot at a quadratic κ of 0.86, and a foundation-model backbone transfers best, matching the premise on which the RETFound baseline (Experiment II) was chosen. The persistent gap between QWK and balanced accuracy is the same rare-class weakness this report has tracked throughout: minority grades (1, 3, 4) remain under-recalled off-domain, so the strong κ reflects correct ordering more than per-class sensitivity. The few-shot curves suggest the cheapest route to a deployable cross-site grader is not retraining but a small labelled calibration set fitted on frozen foundation-model features.

L. Limitations

IDRiD here is an augmented mirror (1,239/309 rather than the standard 413/103), so near-duplicate images can straddle its support and test splits and its *few-shot* numbers should be read as optimistic; the zero-shot numbers, which never fit on the support pool, are unaffected. Only one fully clean external set (APTOS) is used, and zero-shot scores are single point estimates without bootstrap confidence intervals. The study is grade-only: lesion transfer is out of scope because external datasets do not share the seven-lesion scheme. Few-shot uses a linear probe, not full fine-tuning, and the transfer is CFP-only—no public ultra-widefield or OCT grading set exists to test the UWF/OCT models in kind, which is why the UWF model appears only as a control.

M. Conclusion

Evaluated on data they had never seen, the MMRDR-trained CFP models generalise: both grade APTOS zero-shot at QWK ≈ 0.85 –0.86, the RETFound foundation model transfers best on the clean benchmark and recovers most from a handful of labelled examples on the shifted one, and a wrong-modality control transfers measurably worse. The experiment closes the report by showing that the representations it built are not dataset-bound, and that a foundation-model backbone plus a light few-shot calibration is the most promising path to cross-site DR grading.

XIV. CONCLUSION

This report documented ten multi-task deep-learning experiments on the MMRDR dataset, each jointly performing DR severity grading and seven-way lesion detection, and a concluding cross-dataset generalization study (Experiment XI), eleven in total. The first three establish modality baselines.

On color fundus photography, the multi-scale dual-branch convolutional network (Experiment I) reached 82.91% validation grade accuracy, while the fine-tuned RETFound foundation model (Experiment II) reached 80.94% on the dataset’s official held-out test split, reproducing three of the four published baseline metrics within their 95% confidence intervals. These two numbers are close in magnitude but are *not* directly comparable: Experiment II is the only experiment evaluated on a genuine held-out test partition, whereas Experiment I reports validation accuracy on a random split. Experiment II also illustrates a methodological point that generalises beyond this dataset—an earlier attempt reported 66.50%, which an audit traced not to the method but to a silent weight-loading failure in which only 4 of 294 tensors transferred, and to a layer-freezing routine that never executed. On the more challenging ultra-widefield modality, the feature-pyramid network with spatial attention (Experiment III) reached 74.48%.

The remaining five experiments form a focused thread on the ultra-widefield modality: making the model sensitive to the rare, sight-threatening lesions of proliferative DR (NV, VH, RD), attacking the problem from four angles—loss weighting, loss shape, representation, and data sampling. Experiment IV re-weights the lesion loss by combined statistical rarity and clinical risk, lifting recall on the highest-risk lesions (RD 0.79, VH 0.77, NV 0.68) at the intended cost of common-lesion recall. Experiment V replaces the objective with an asymmetric, false-negative-penalised focal loss, reaching a quadratic κ of 0.92 and lesion AUC of 0.95. Experiment VI shapes the feature space with a two-depth supervised contrastive auxiliary. Experiment VII learns the multi-task weights via homoscedastic uncertainty and reaches the study’s best rare-lesion detection (lesion F1 up to 0.778, AUROC up to 0.961; rare recall NV/VH/RD = 0.89/0.95/0.96), though its lesion weight saturated at the numerical clamp. Experiment VIII holds the network fixed and adapts the training *sampler* with a recall-driven curriculum, pushing RD recall to 1.0 and reaching $\kappa = 0.923$ on a stratified split. Two further experiments probe the architecture and the data supply. Experiment IX is a controlled, capacity-matched ablation on a held-out test split that finds a dedicated rare-lesion head does not beat—and slightly harms—a shared head (0.816 vs. 0.890 mean rare recall), indicating head topology is not the bottleneck. Experiment X completes the generative-augmentation study: a 64px conditional-diffusion generator’s synthetic rare-lesion images do *not* improve downstream rare-class recall over a real-only baseline (0.833 vs. 0.878) or over oversampling, and a very high synthetic-vs-real FID (≈ 247) localises the bottleneck to generation fidelity—a clean negative result that motivates a higher-resolution generator. Finally, Experiment XI steps outside MMRDR entirely and tests cross-dataset generalization: applied to unseen external fundus datasets (APTOS 2019 and IDRiD), the two CFP models grade APTOS zero-shot at a quadratic κ of ≈ 0.85 –0.86, the RETFound foundation model transfers best and recovers most from a few labelled examples, and a wrong-modality UWF control transfers measurably worse—evidence that the learned representations are not dataset-bound. Across the ten trained experiments, focal-family objectives stabilised

TABLE XXV
SUMMARY OF THE TEN EXPERIMENTS

Exp.	Model / lever (modality)	Grade acc.
I	Multi-scale dual-branch CNN (CFP)	0.8291
II	RETFound ViT-L/16 fine-tune (CFP)	0.8094 *
III	FPN + spatial attention (UWF)	0.7448
IV	Clinically-weighted lesion loss (UWF)	0.7650
V	Asymmetric FN-penalised loss (UWF)	0.7578 ^a
VI	Supervised contrastive auxiliary (UWF)	0.7645 ^b
VII	Learned uncertainty weighting (UWF)	0.7554 ^c
VIII	Recall-driven resampling (UWF)	— ^d
IX	Shared vs. separate heads (UWF)	— ^e
X	Synthetic generation (UWF)	— ^f

*Official held-out test split (not validation); paper 0.822.
^aSelected ckpt, ep 16; peak 0.7770. ^bPlateau peak, ep 28–37.
^cSelected ckpt, ep 21; peak 0.7645. ^d $\kappa = 0.923$ (acc. not logged).
^eTest split; best macro-F1 0.707, $\kappa = 0.914$. ^fVerdict: synthetic does not beat real-only rare recall (0.833 vs. 0.878).

TABLE XXVI
UWF RARE-LESION EXPERIMENTS — VALIDATION METRICS

Exp. (lever)	Grade κ	Lesion F1	Lesion AUC	RD rec.
IV (clinical wts)	—	0.56	—	0.79
V (asym. loss)	0.9208	0.7151	0.9504	—
VI (contrastive)	0.9223	0.7390	0.9523	—
VII (uncertainty)	0.9211	0.7779	0.9605	0.958
VIII (resampling) [†]	0.9234	0.7088	0.9455	1.00

Best validation value per column. Lesion F1/AUC macro-averaged; “—” = not computed. RD rec. = retinal-detach. recall. [†]Stratified split—not directly comparable to the other rows.

training under class imbalance, and Grad-CAM confirmed that predictions were driven by clinically plausible retinal regions. Tables XXV and XXVI summarise the comparison. Future work includes multi-seed, same-split comparisons with significance testing, a wider uncertainty clamp, higher-resolution foundation-model fine-tuning, calibrating the risk weights to a clinical utility function, running the generative-augmentation study, and unified multimodal training.

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